



UNIVERSITY OF AMSTERDAM
Amsterdam Business School

Data Science in Online Marketing

Online Advertising – Introduction

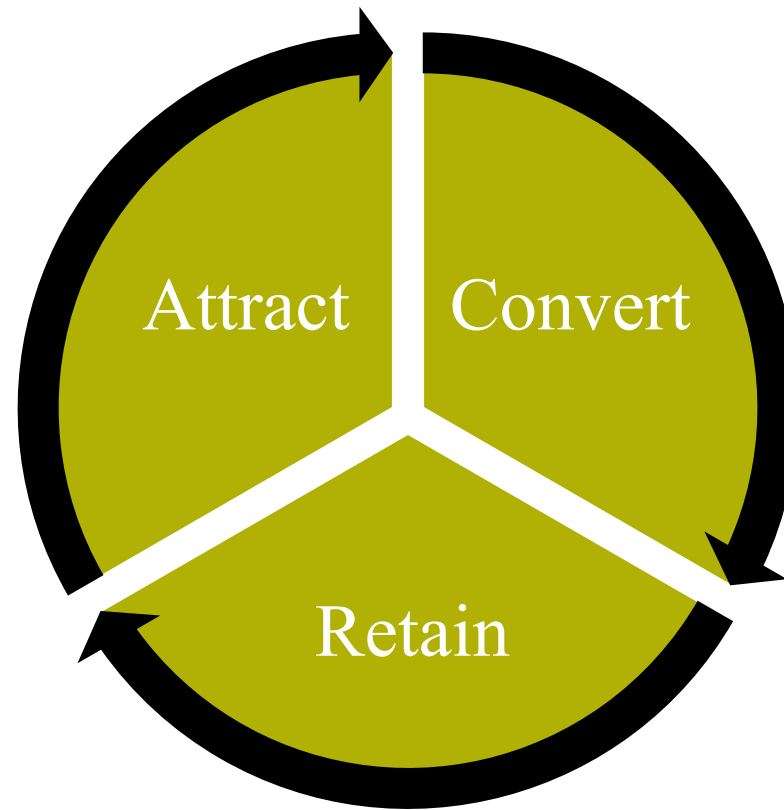
Maarten Soomer



Course Outline

Date	Topic	Assignment
3 april	Recommenders: Content Based	
10 april	Recommenders: Collaborative Filtering	Yes
17 april	Recommenders: Guest Lecture Bol.com Online advertising: Introduction	
24 april	Online advertising: Reinforcement Learning	
1 may	Online advertising: CTR prediction	Yes
15 may	Online advertising: Media Mix Modeling & Conversion Attribution	
22 may	Revision	
29 may	Digital exam on location	

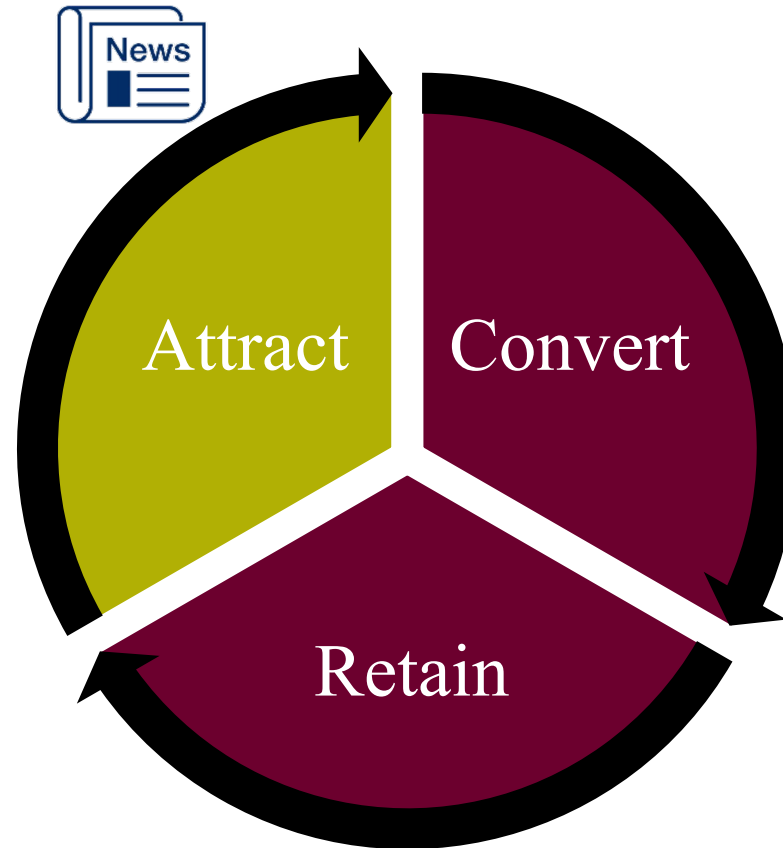
E-commerce sales cycle



E-commerce sales cycle

Marketing & advertising to attract (new) customers:

- Direct Cost involved: Marketing Investment
- Large **online advertising** budgets: How to spend effectively?

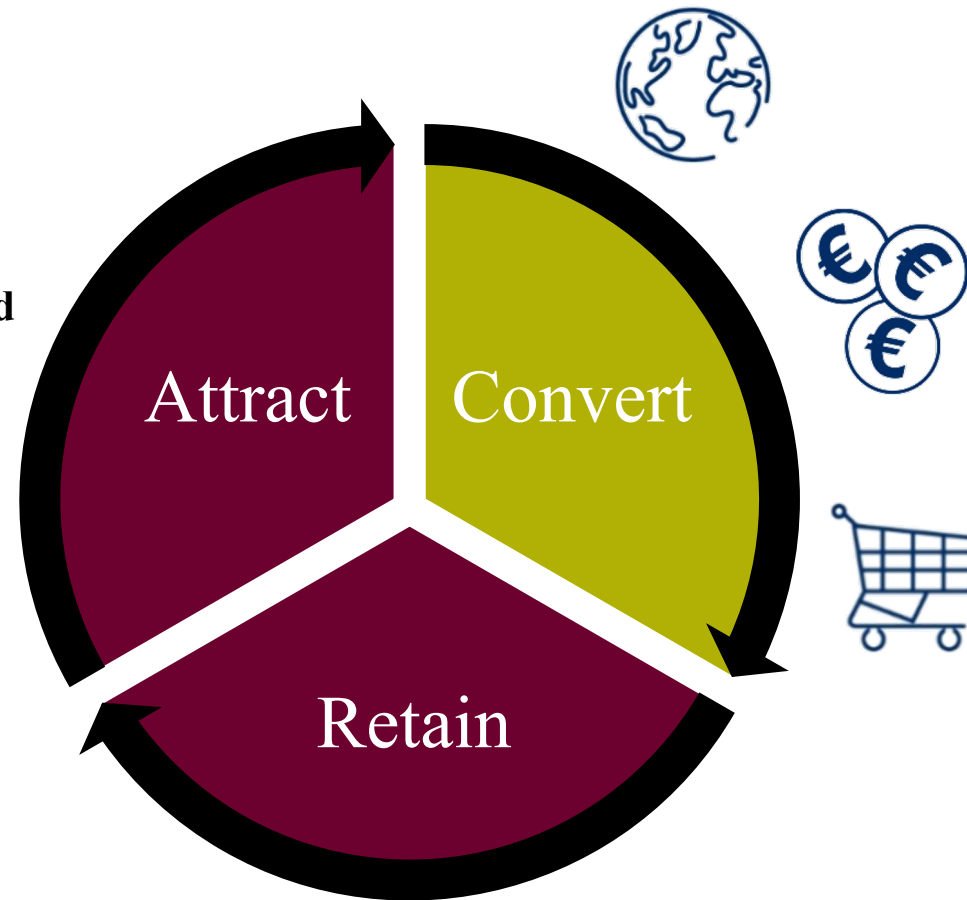


E-commerce sales cycle

Convert visitors of e-commerce site:

- Relevant Product Offering
- **Relevant, personalized & optimized website**
- Competitive Price

Make sure marketing investment is not wasted by a bad user experience



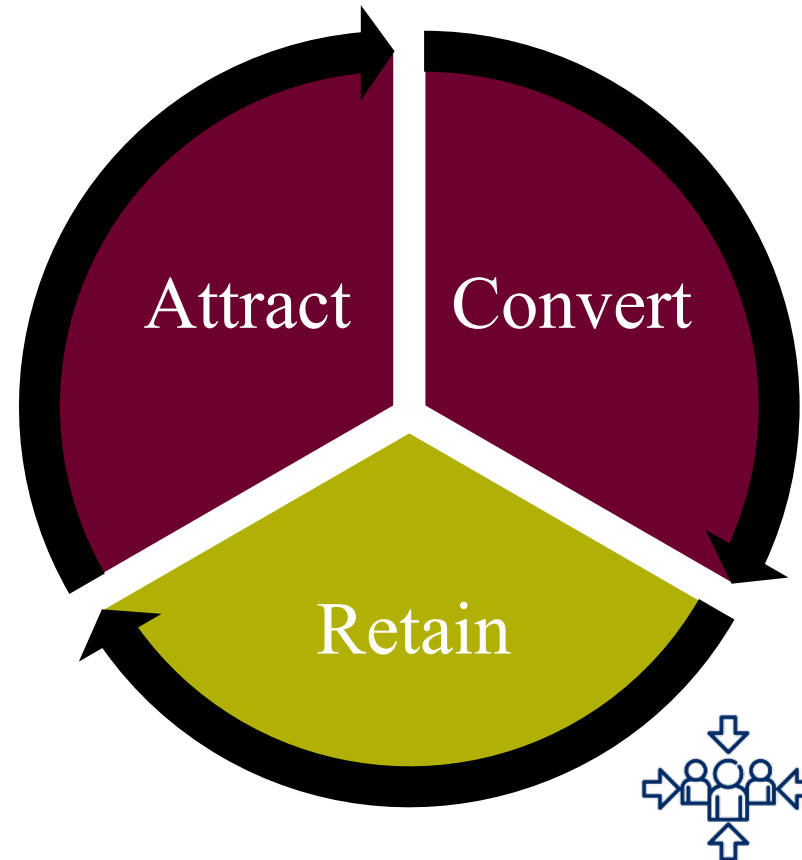
E-commerce sales cycle

Retain existing customers:

- Excellent Service
- **Relevant, personalized recommendations** and communication

Increase customer lifetime value and minimize advertising investment need to re-activate customers.

Use your knowledge of the customer.





Data Science in Online Marketing course

Get a deep dive on concepts and algorithms used in online marketing:

Online Advertising

- **CTR prediction**
- **Media Mix Modeling & Conversion Attribution**



Recommenders

- Content Based
- Collaborative Filtering



amazon

NETFLIX





UNIVERSITY OF AMSTERDAM
Amsterdam Business School

Online Advertising



“Half the money I spend on advertising is wasted; the trouble is I don’t know which half”

Johan Wanamaker (1838 – 1922)

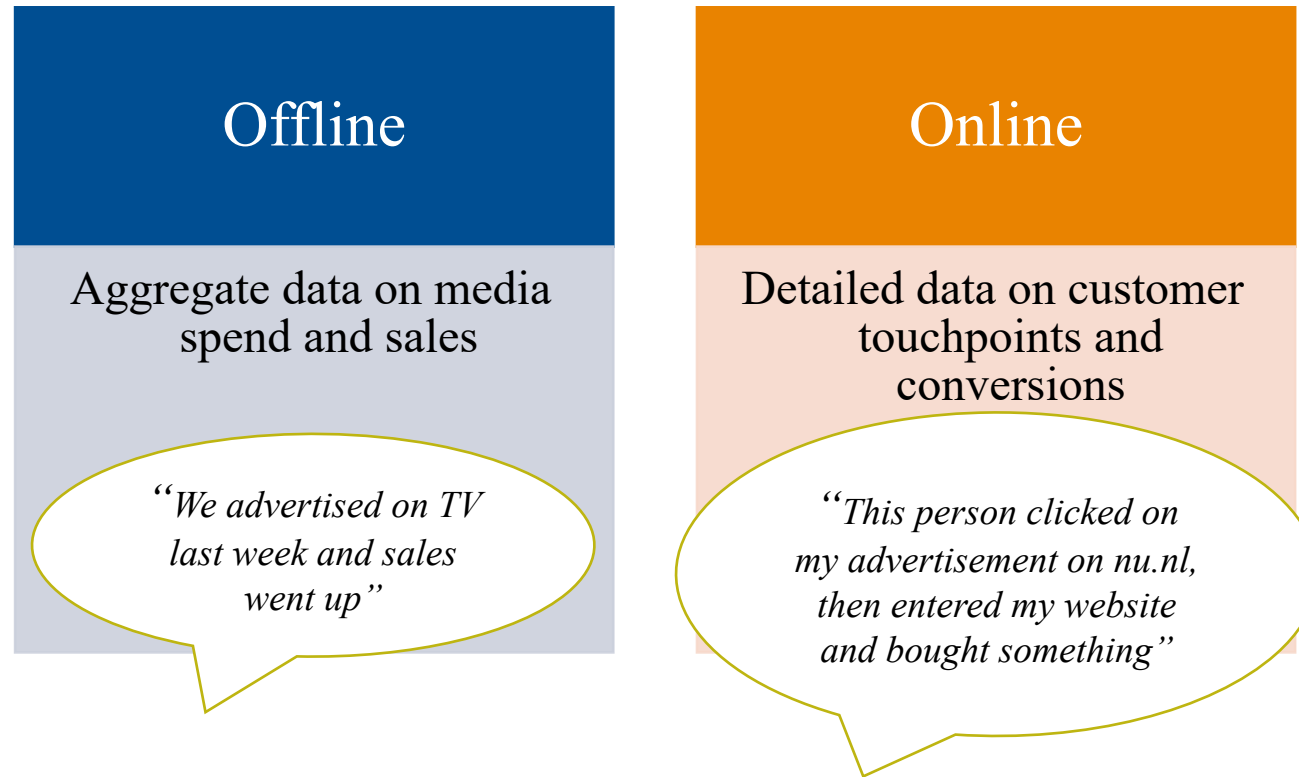
Father of modern advertising and a “pioneer in marketing.”

Fast Forward to the online era

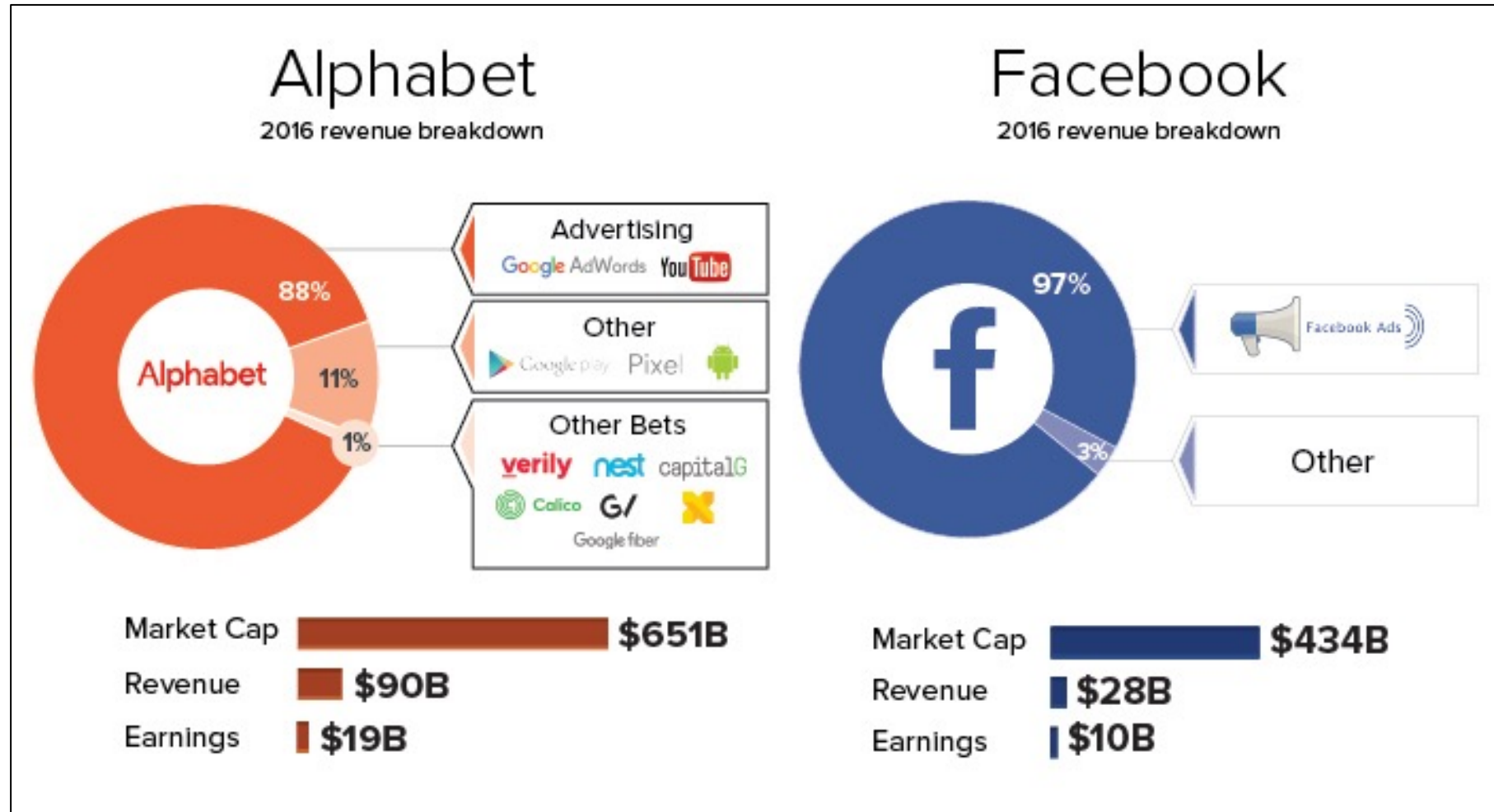




Offline vs online



How do Google & Facebook earn money?



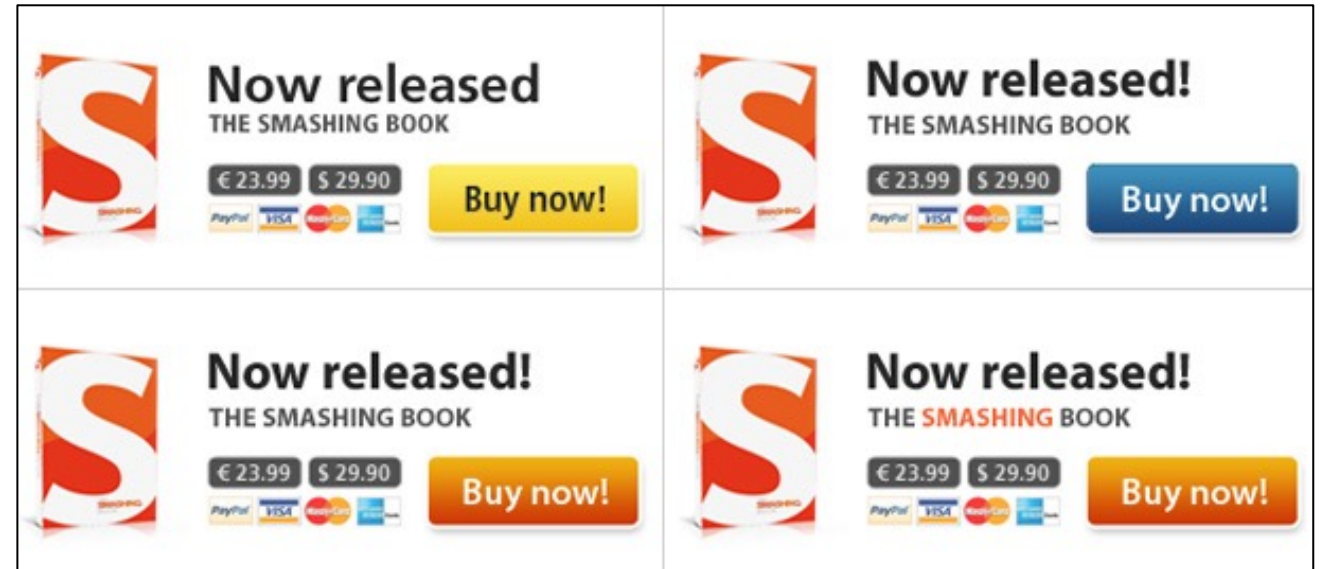


Online advertising

- Targeted
- Real time
- Interactive
- Measurable
- Low entry cost

(Online) Advertising Challenges

- What message?
- When to advertise?
- What should my advertisement look like?
- Who to target?
- Which channels?
- How much money to spend?



**The right advertisement for the right person against the right price
at the right time & channel**



Online Advertising - Outline

- Funnel & Metrics
- Systems & Data
- Online advertising channels:
 - Social Media
 - Search Engines
 - Display (Real Time Bidding)
- Multi channel:
 - Conversion attribution



Online Advertising - Outline

- **Funnel & Metrics**
- **Systems & Data**
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Online Advertising Funnel

Target Audience

- People that I want to see my ad

Impressions

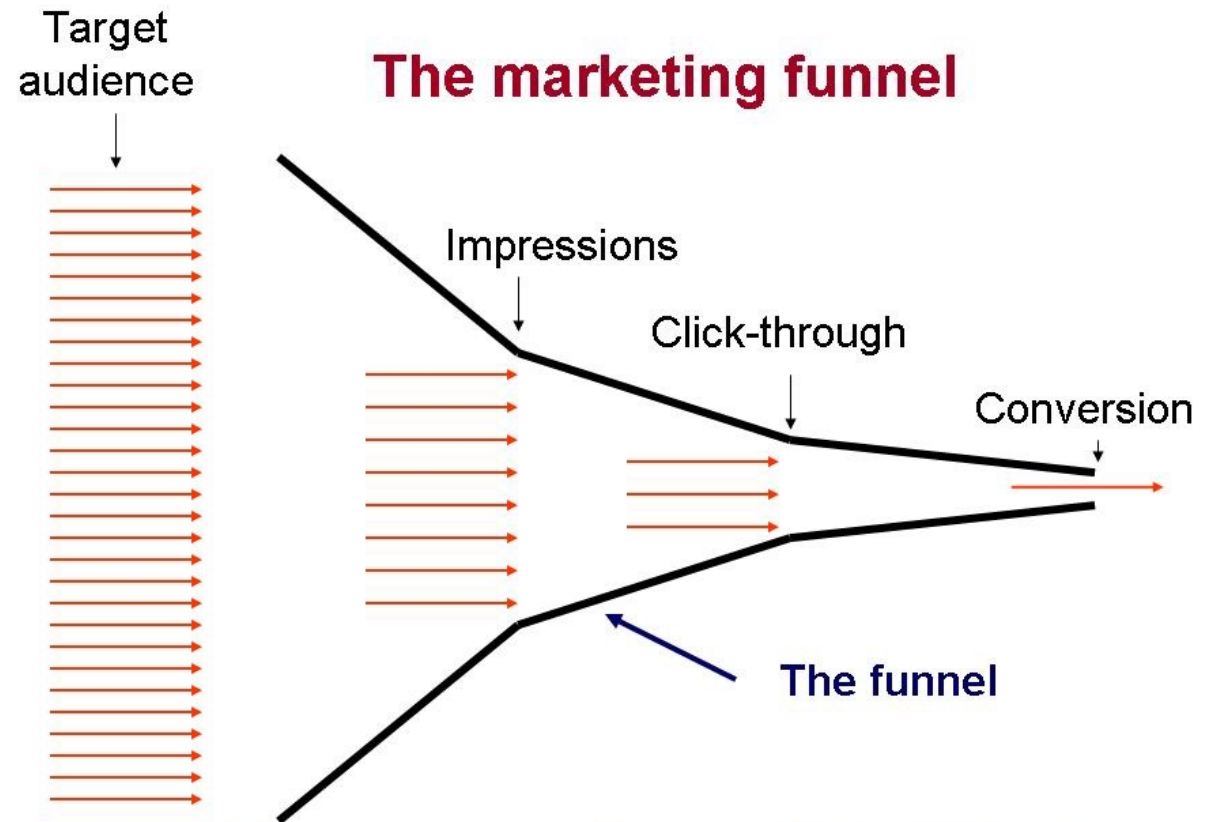
- People who actually see my ad

Click-through

- People who click on my ad
(resulting in a website visit)

Conversion

- People who buy a product on my website



Targeting

Contextual

- **The content of the website matches the content of the ad:**

KLM ad on travel website

Coffee ad in the morning

Behavioral

Privacy
sensitive

- **The ad is shown based on the behaviour and other data of the website visitor**

Heineken ad after visiting sports websites

Zalando ad after visiting Zalando website
(**retargeting**)

Ajax ad only in Amsterdam region
(**geographical targeting**)

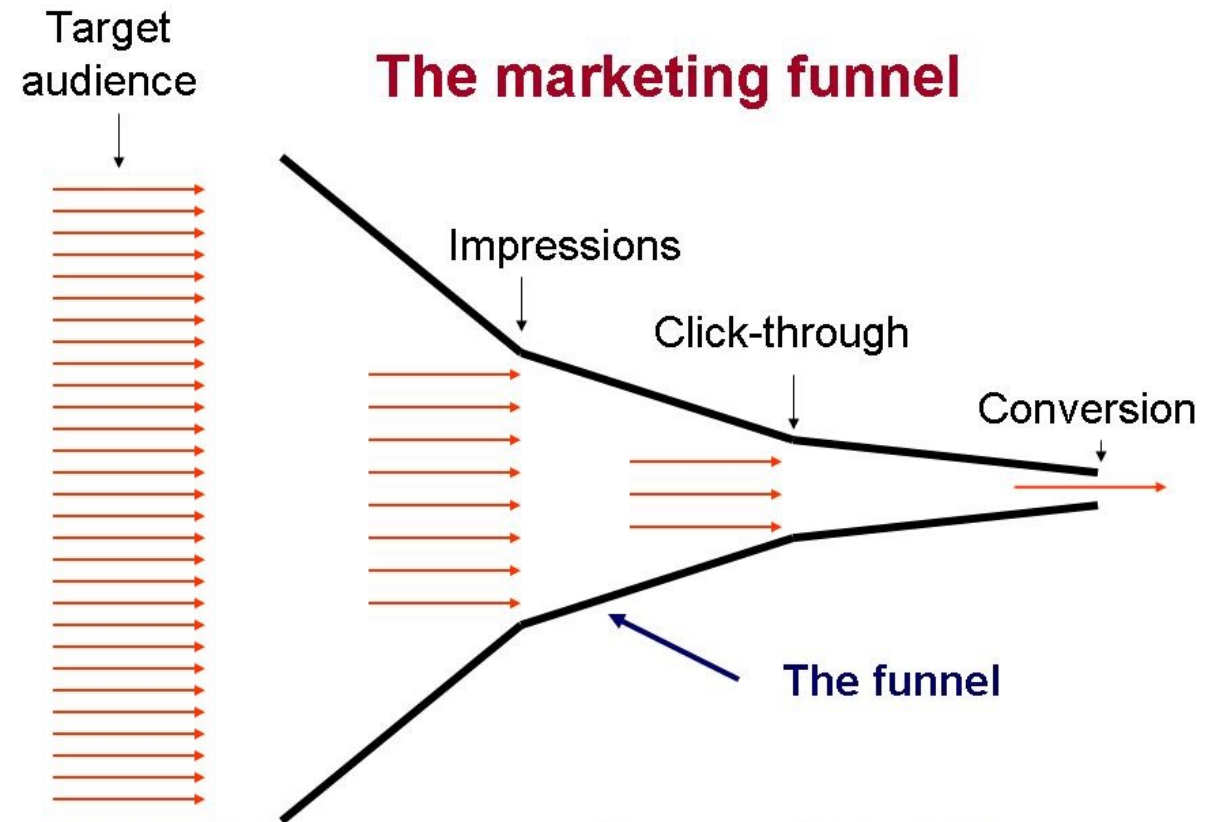
Online Advertising Funnel – Performance metrics

Click-through rate (CTR) =

$$\frac{\# \text{ click throughs}}{\# \text{ impressions}}$$

Conversion rate (CR) =

$$\frac{\text{conversions}}{\# \text{ click throughs}}$$



Online Advertising Funnel – Performance metrics

Cost metrics:

- Cost per impression

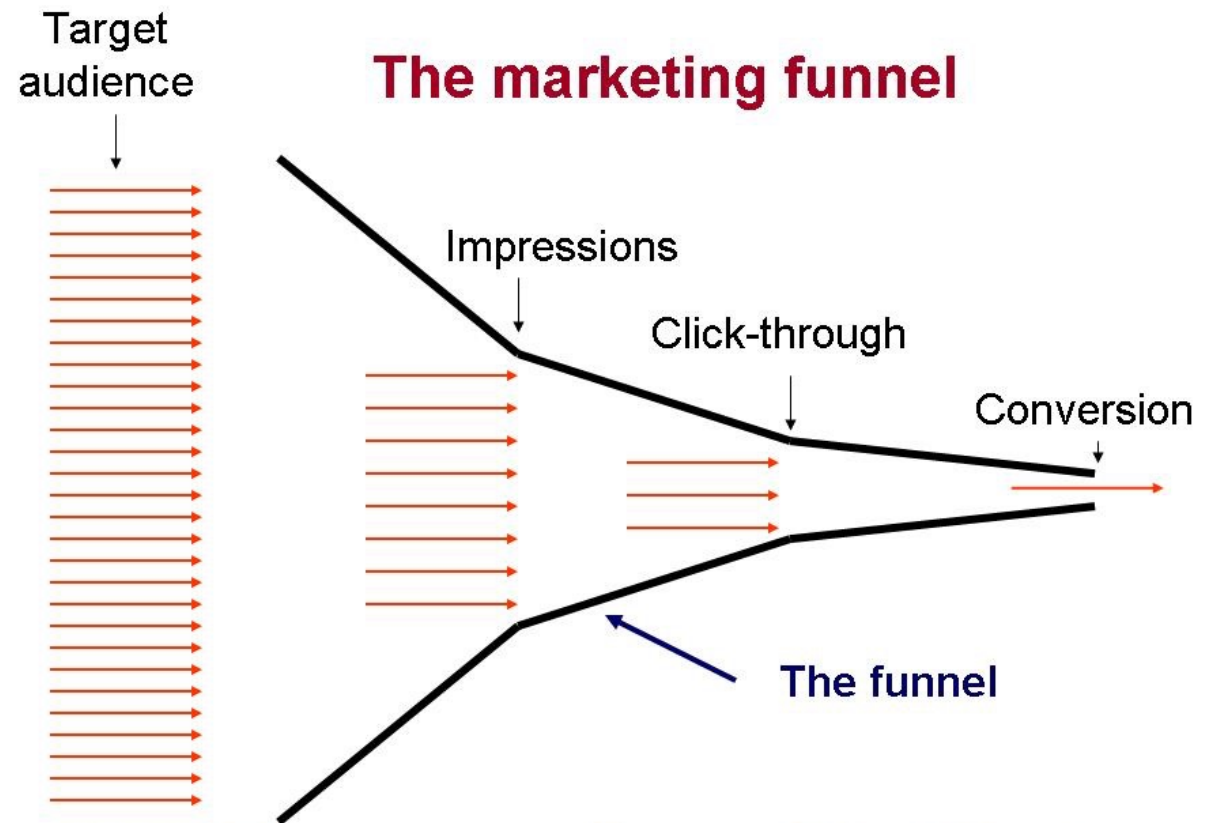
(CPM = cost per mille)

- Cost per click (CPC)
- Cost per conversion (CPA)

Efficiency:

- Return on ad spend (ROAS)

$$= \frac{\text{conversion revenue}}{\text{ad spend}}$$

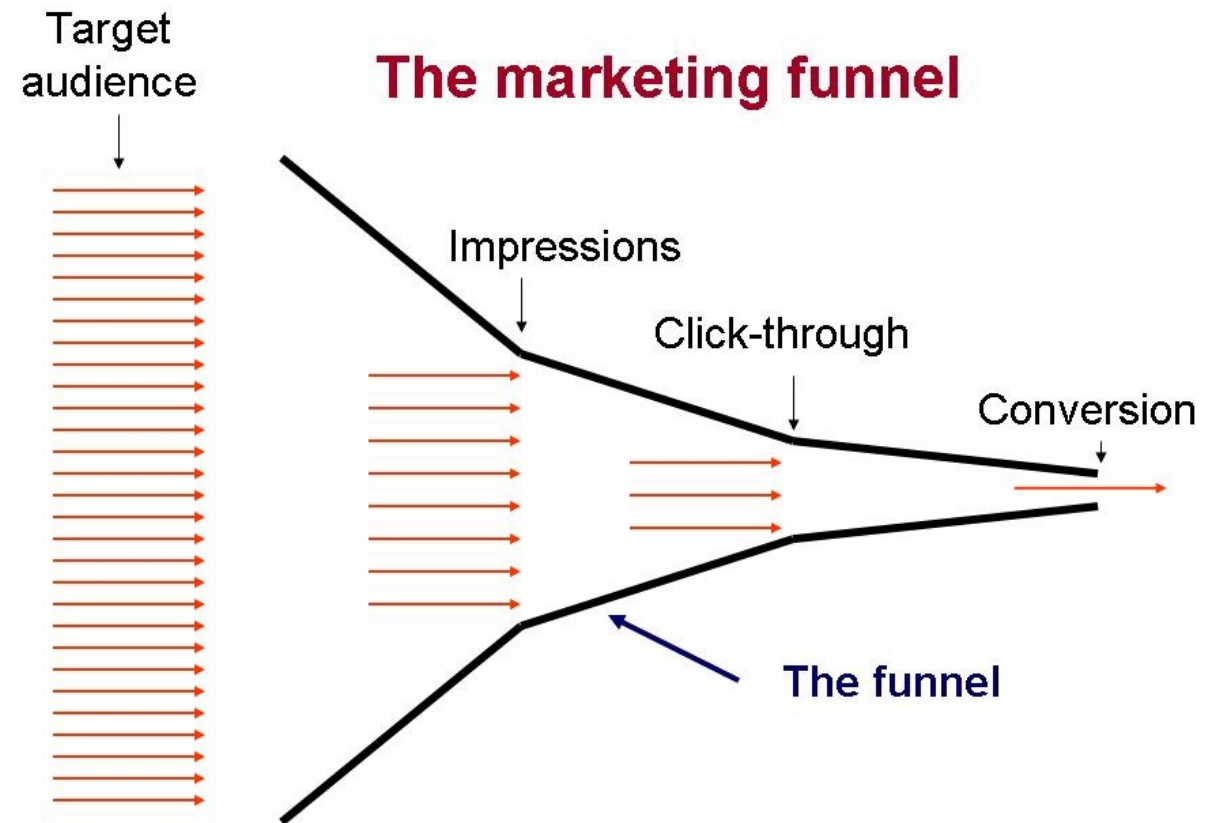


Online Advertising Funnel – Example

Ad on CNN website:

- Cost: \$5 per 1.000 impressions (CPM)
- 10.000 impressions
- 1.000 clicks
- 2 conversions

- CTR =
- CR =
- CPA =



Online Advertising Funnel – Example

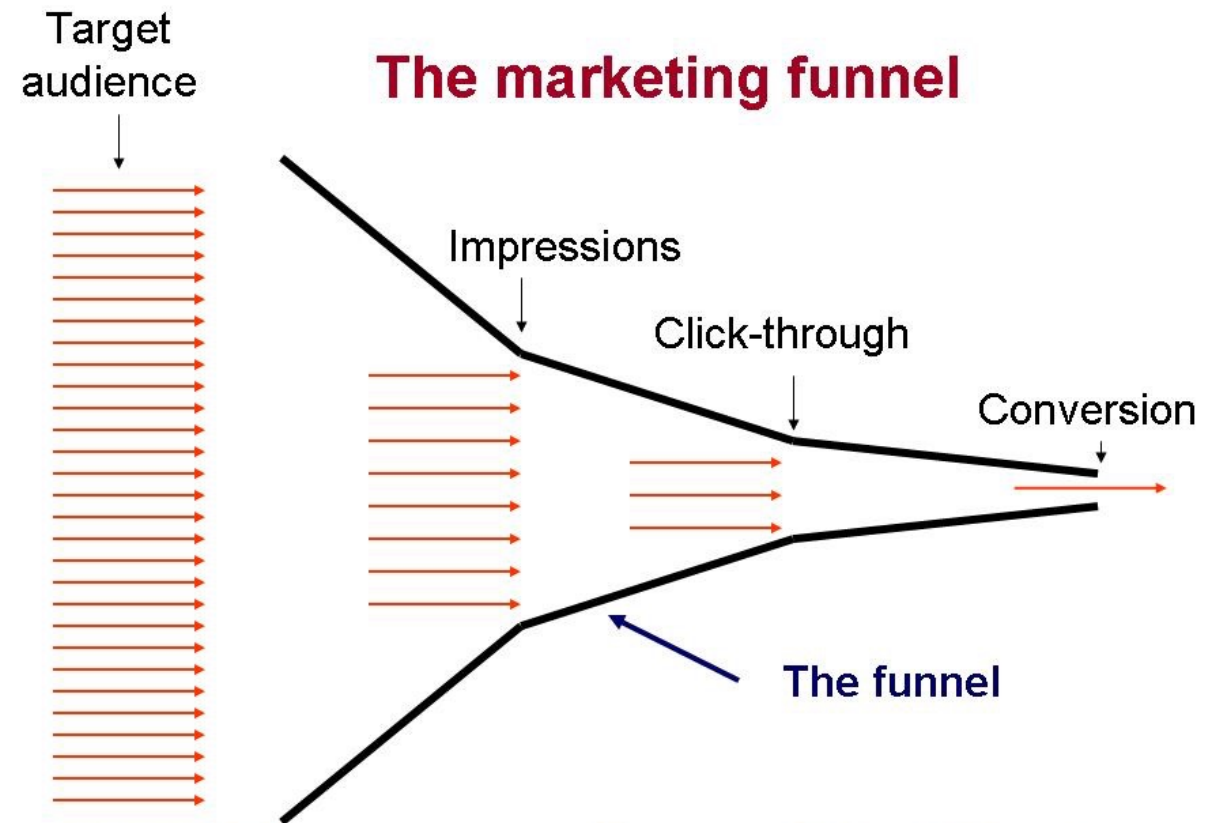
Ad on CNN website:

- Cost: \$5 per 1.000 impressions (CPM)
- 10.000 impressions
- 1.000 clicks
- 2 conversions

- $CTR = \frac{1.000}{10.000} = 10\%$

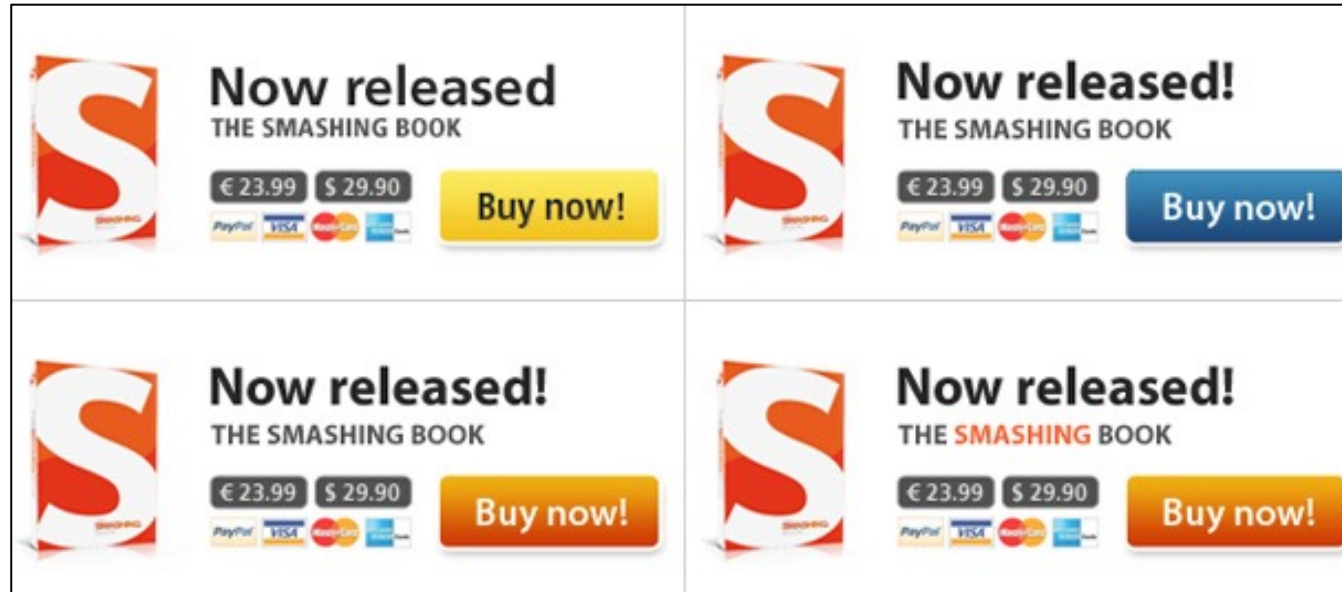
- $CR =$

- $CPA =$



Advertisement Layout Optimization

Which version of my advertisement will have the highest CTR?



How to measure/predict/compare CTR of the different versions?

Online Advertising Funnel – Example

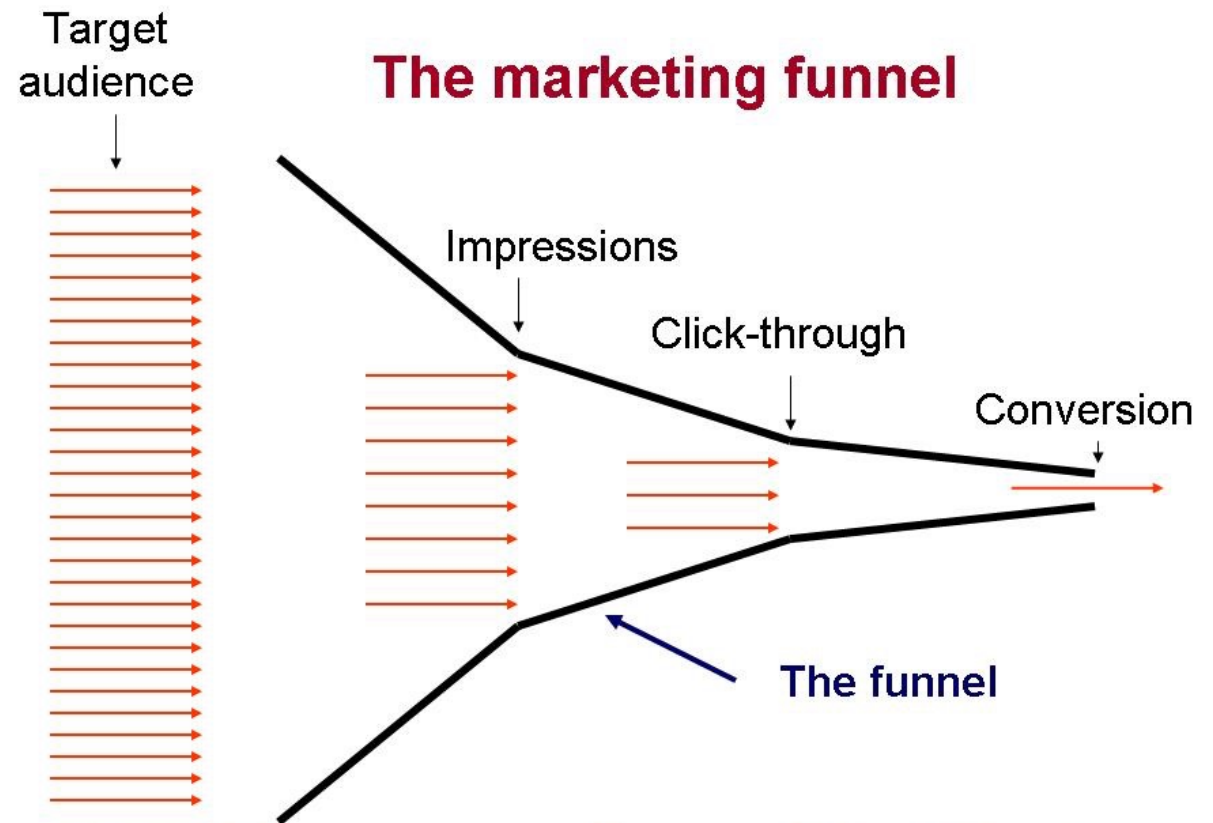
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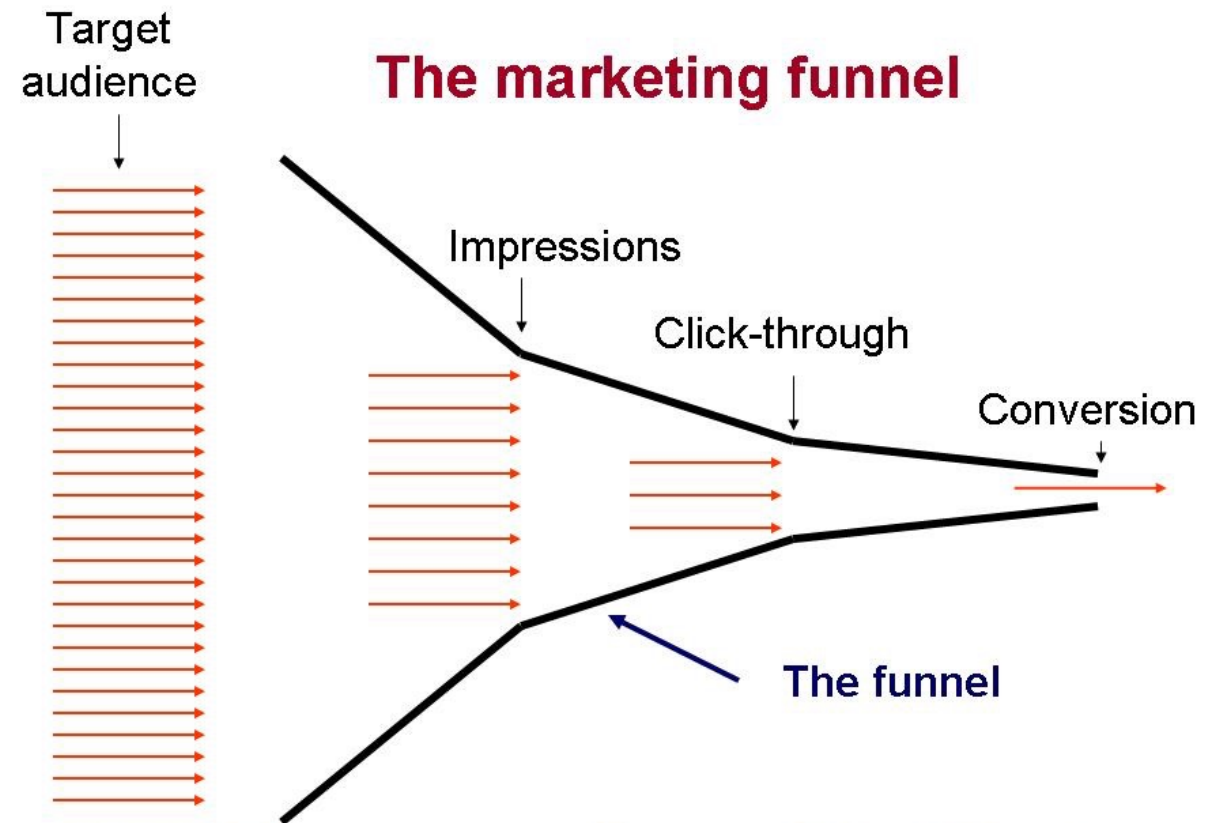
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- Cost: \$5 per 1.000 impressions (CPM)
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- $CR = \frac{2}{1.000} = 0.2\%$

- $CPA =$



Online Advertising Funnel – Example

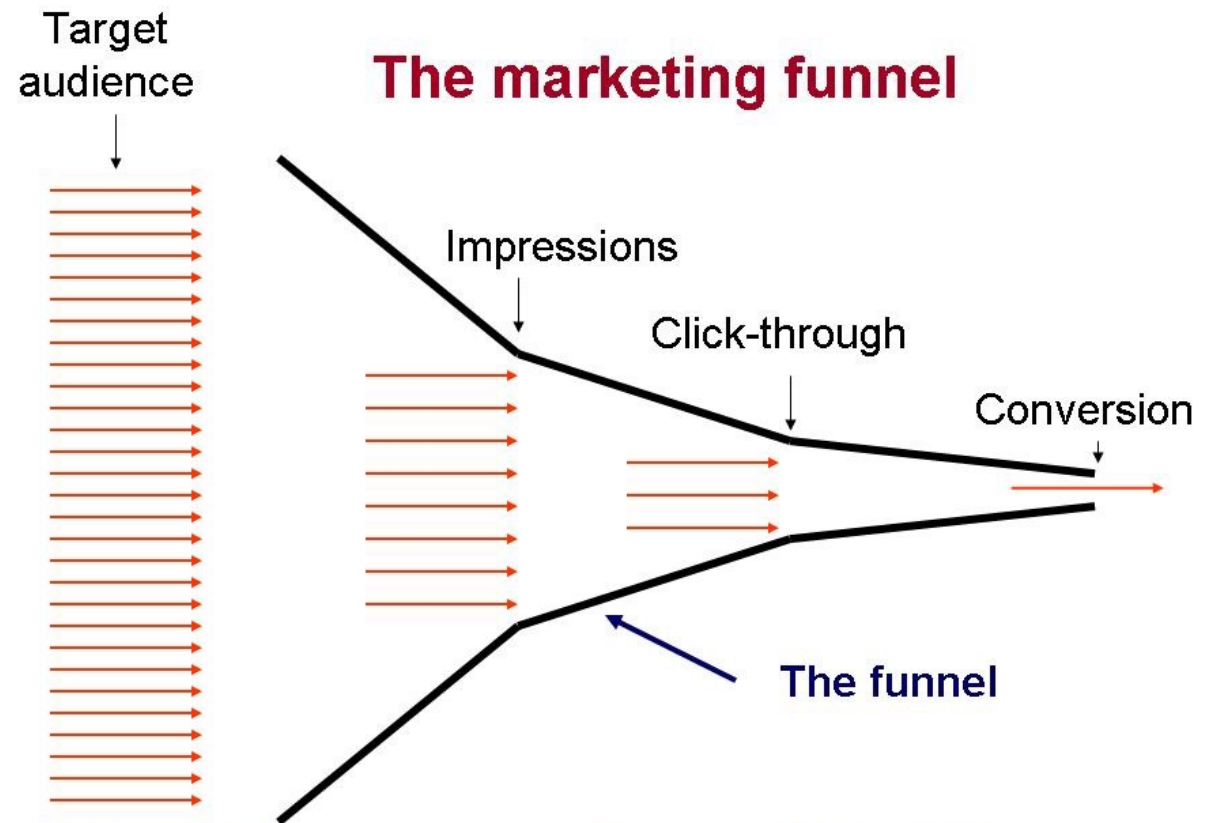
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- Cost: \$5 per 1.000 impressions (CPM)
- 10.000 impressions
- 1.000 clicks
- 2 conversions

- $CTR = \frac{1.000}{10.000} = 10\%$

- $CR = \frac{2}{1.000} = 0.2\%$

- $CPA = \frac{\$5/1.000 * 10.000}{2} = \25

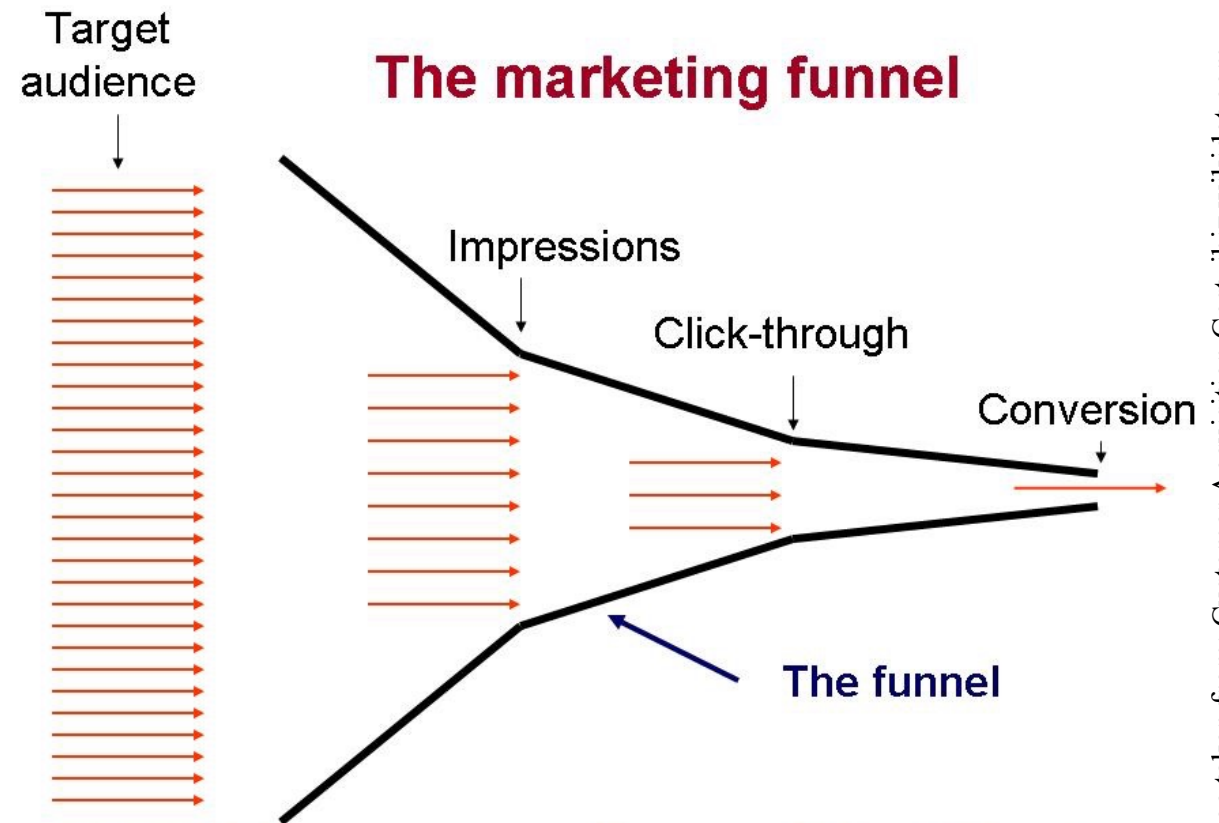


Online Advertising Funnel – Optimization

- Which budget (ad spend) will give the maximum conversion revenue/profit?
 - $\text{Profit} = \text{conversion margin} - \text{ad spend}$
- How much conversions/revenue will I get given a fixed budget or a fixed (max) CPA?

To answer these questions:

What is the relation between the ad spend and number of conversions?





Online Advertising - Outline

- Funnel & Metrics
- **Systems & Data**
- Online advertising channels:
 - ☐ Social Media
 - ☐ Search Engines
 - ☐ Display (Real Time Bidding)
- Multi channel:
 - ☐ Conversion attribution



Systems & Data - Basics

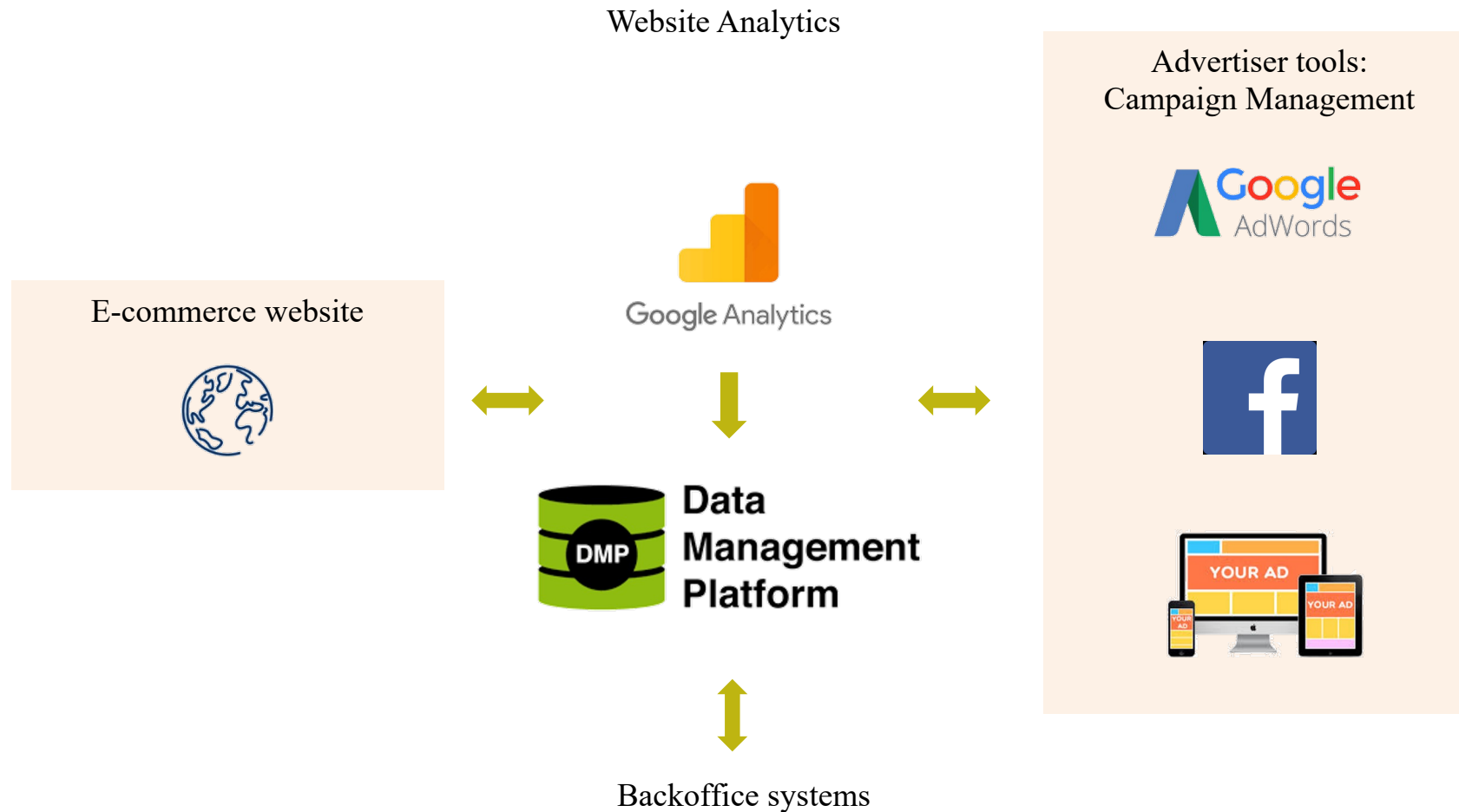
■ IP address

- ☐ Address of your device
- ☐ Internet doesn't work on your device without it

■ Cookie

- ☐ Tiny text file in your web browser that stores information
- ☐ It personalizes your browsing experience (e.g. login information, user preferences, shopping cart)
- ☐ 1st party Cookies: only used on the website itself
- ☐ 3rd party Cookies: Cookie from another party on a website, can be used to track across websites

Systems & Data – E-commerce (advertiser)

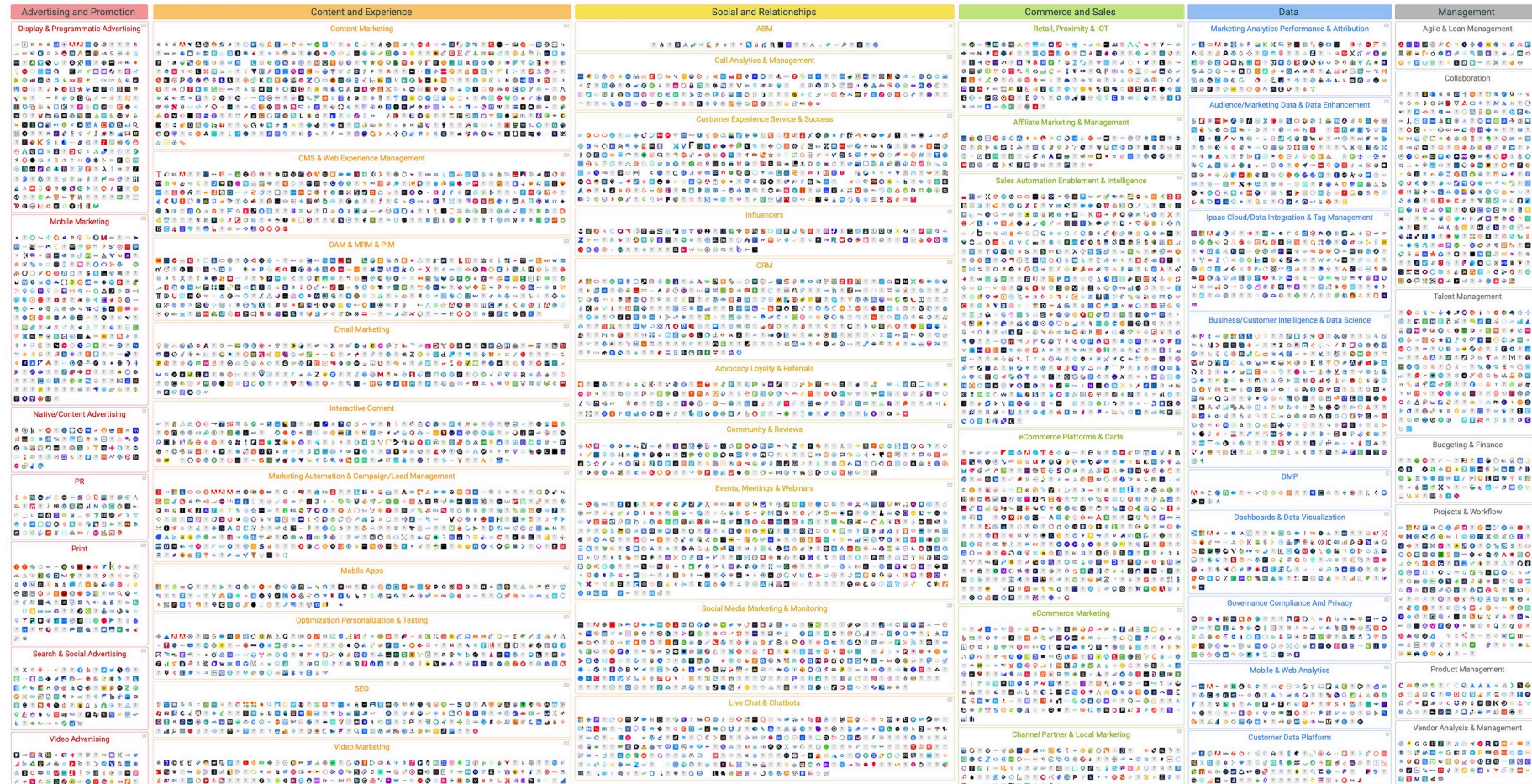




Systems & Data

MartechMap an initiative by  chiefmartec &  MartechTribe

2023 Marketing Technology Landscape May 2023



visit martechmap.com to search, sort & filter



Data – E-commerce (advertiser)

- Touch-point data

- ☐ When and where did this IP-address see or click on one of my ads?
 - As soon as you visit a website from one of my ads

- Session data

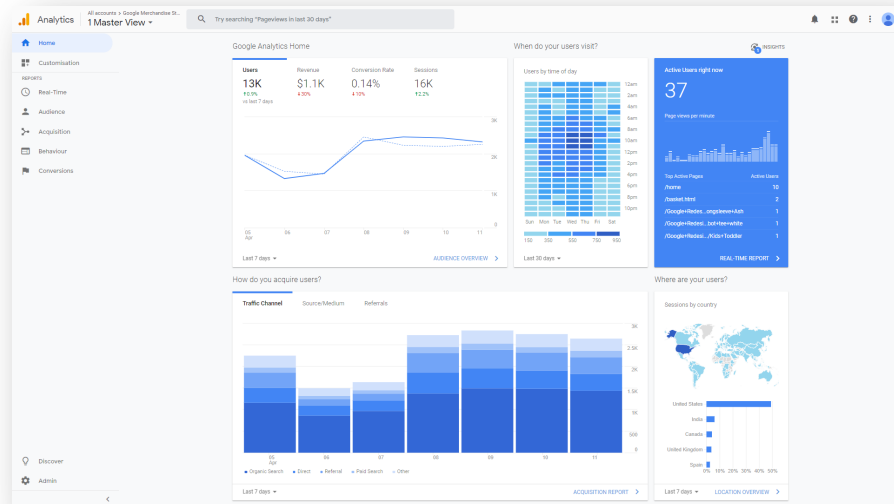
- ☐ Everything you do when you visit my website
- ☐ What pages you see, how long, what you search for, ...
 - As soon as you visit my website

- Personal data

- ☐ E-mail, street address, purchase history, etc.
 - Only when you give your data to me (usually: login or sale)

Web Analytics

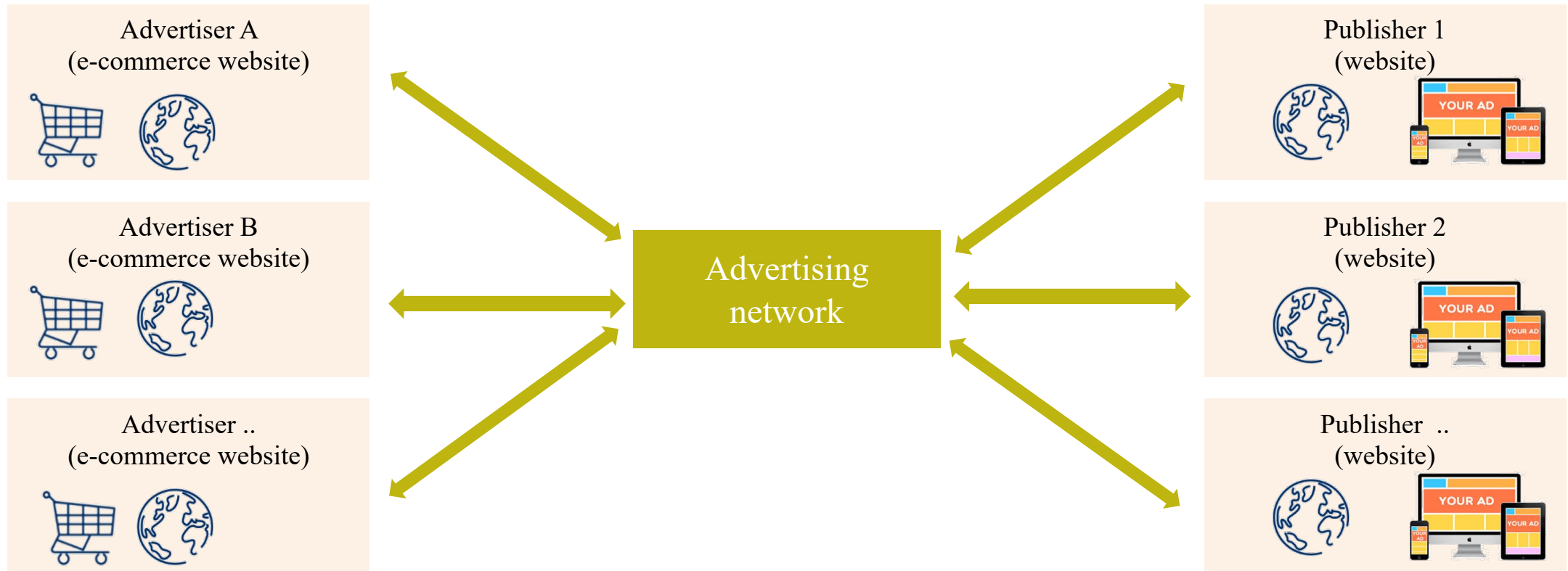
- A web analytics system stores website session data
- Google Analytics is the most known and used



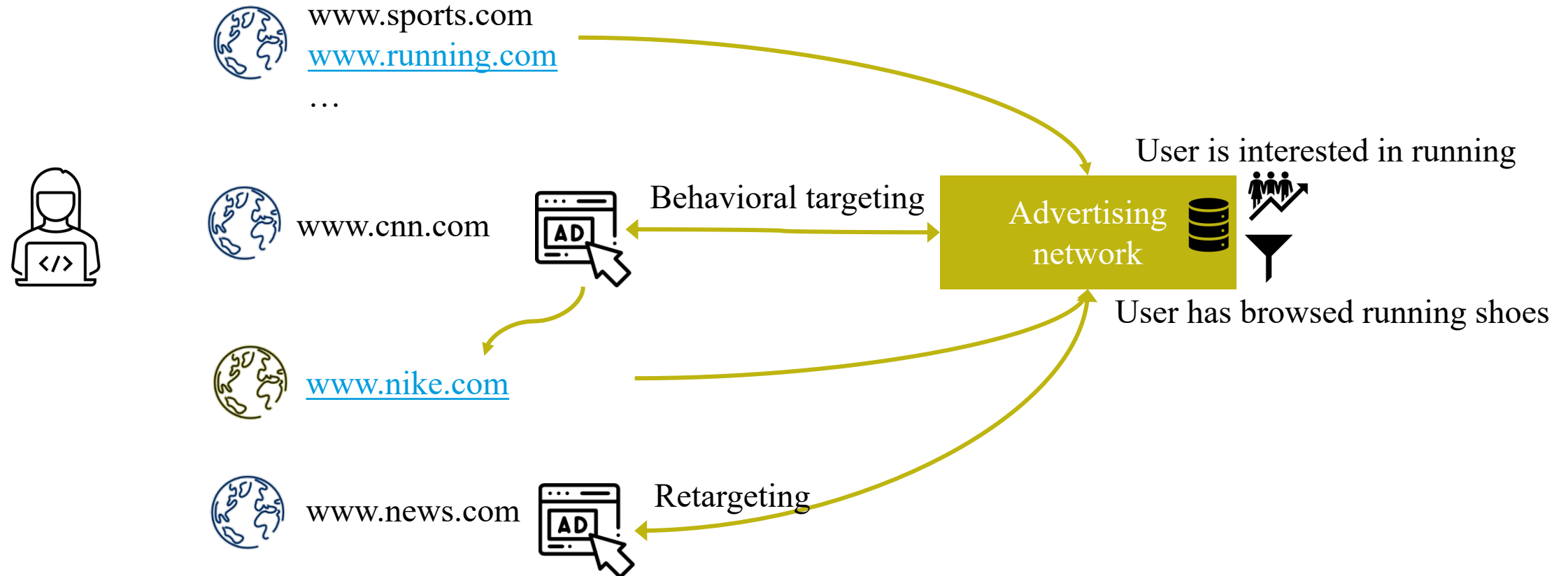
Demo:

<https://analytics.google.com/analytics/web/demoAccount?appstate=/p213025502>

Systems & Data –Advertising Networks



Third party Cookies



Third party Cookies



The Verge / Tech / Reviews / Science / Entertainment / AI / More +

Google will turn off third-party tracking for some Chrome users soon



Illustration: The Verge

/ Google's plan to implement a new Tracking Protection feature in Chrome begins in January with the intention to completely disable third-party cookies in the second half of 2024.

By [Umar Shakir](#), a news writer fond of the electric vehicle lifestyle and things that plug in via USB-C. He spent over 15 years in IT support before joining The Verge.

Dec 14, 2023, 3:00 PM GMT+1

[Link](#) [Facebook](#) [Twitter](#) | 14 Comments (14 New)

Third party Cookies - Alternatives

Currently alternatives for 3rd party cookies are under development, to allow for some forms of behavioral targeting.

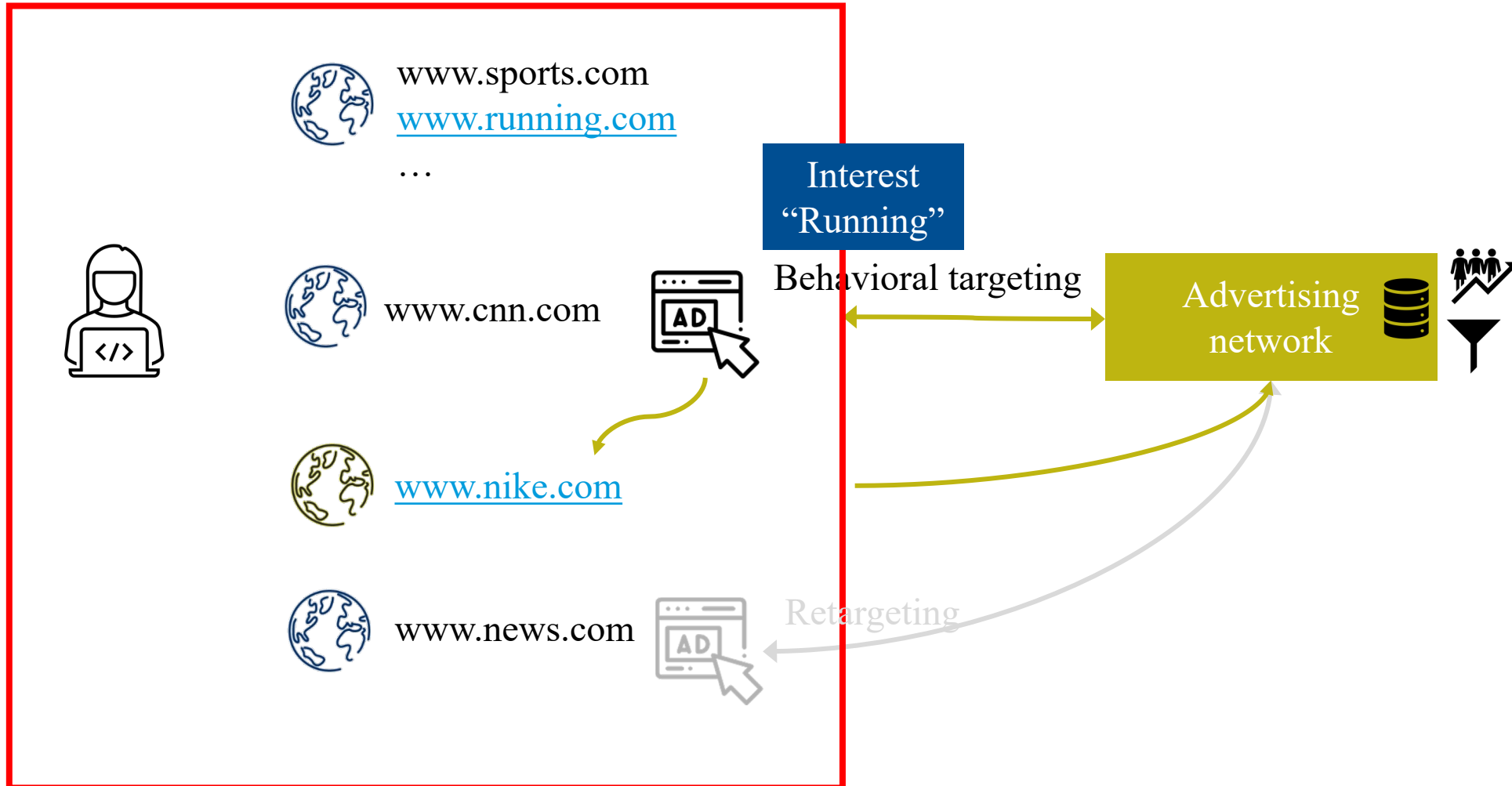
Concepts/Ideas of the alternatives:

- Data about cross-site browsing behaviour cannot be tracked anymore
- **Browser** determines user interests **on device** based on (recent) browsing behavior
 - Interests are learned from user browsing behaviour on device by the browser
 - Interests are not specific enough to identify individual users, no “sensitive” interests are included
 - User has control over interests (e.g., view, turn off) in browser
- Advertising networks can use the interests to target (groups of) users

Further reading:

- <https://blog-google.cdn.ampproject.org/c/s/blog.google/products/chrome/get-know-new-topics-api-privacy-sandbox/amp/>

Third party Cookies - Alternatives



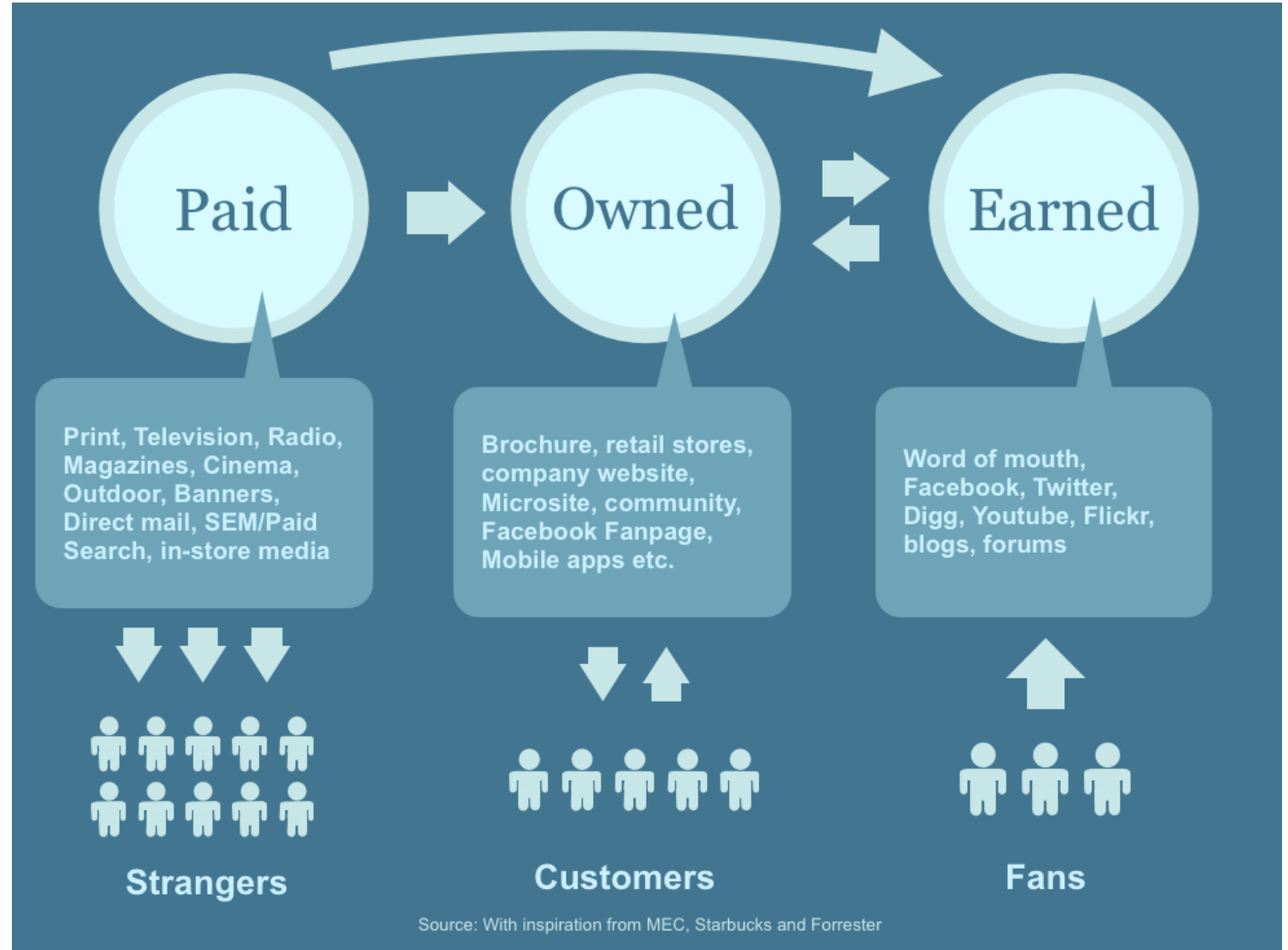
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 - ☐ Conversion attribution



Online channels

A way to categorize online channels:





Social Media



Social Media – Targeting your audience



Location

Country: [?] **United States** x

☒ Everywhere ☐ By State/Province [?] ☐ By City [?]

Demographics

Age: [?] **18** - **30**

☐ Require exact age match [?]

Gender: [?] ☒ All ☐ Men ☐ Women

Likes & interests

Climbing x **Mountaineering** x **Rock Climbing** x

Suggested likes & interests

Connections on Facebook

Connections: [?]

Target users who are connected to:

Enter your Page, Event, Group or App [?]

Target users who are not already connected to:

Enter your Page, Event, Group or App [?]

Target users whose friends are connected to:

Enter your Page, Event, Group or App [?]

Friends of connections:

☒ Show advanced targeting options

Estimated reach

392,140 people

- who live in the **United States**
- between the ages of **18** and **30** inclusive
- who like **climbing**, **mountaineering** or **rock climbing**

Annotations:

Enter the country/country of your targeted prospects. You can target by state/province or city as well. If you run a local business, it's advisable to target your city and surrounding small cities only.

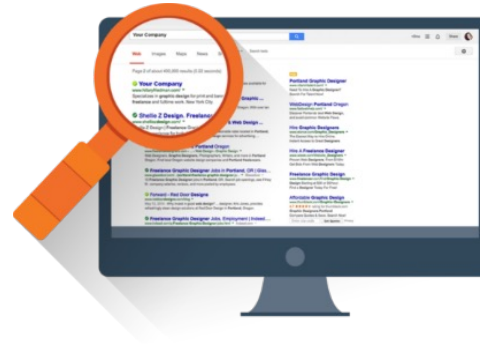
Set your target demographics

Target your audience by interests.

Social targeting. If you want to reach a very specific group of people – example: *all people interested about climbing that aren't fans of your Facebook page.*



Search Engines





hotel amsterdam

All Images Maps Shopping News More Settings Tools

About 307,000,000 results (0.96 seconds)

Hotels in Amsterdam | Up to Half-Price on Hotels | booking.com
Ad www.booking.com/Amsterdam/Hotels ▼
Book your Hotel in Amsterdam online. No reservation costs. Great rates. Get Instant Confirmation. We speak your language. Secure Booking. Motels. No Booking Fees. Best Price Guarantee. Villas. 34+ Million Real Reviews. Apartments. Support in 42 Languages. Types: Hotels, Apartments, Villas.

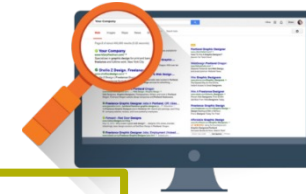
Homes, apartments, & more Book now, stay tonight! Comforts of home, wherever you are	Book apartments and more Bookings instantly confirmed! Instant confirmation, 24/7 support
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Hotels in Amsterdam | 976 Hotels from \$55 per Night
Ad www.trivago.com/Hotels/Amsterdam ▼
Find Your Ideal Hotel in Amsterdam. Compare Prices and Save on your Next Stay! trivago™ Shows Great Offers from Different Websites - All in One Place. Over 1 Million Hotels. Save Time and Money.
[3* Amsterdam Hotels](#) · [Top Amsterdam Hotels](#) · [Last Minute Hotels](#) · [Hotels from \\$55](#)
[Cheap Hotels](#) - from US\$ 50,00/night - Compare Prices · [More](#) ▼

Best Hotels in Amsterdam | Up to Half-Price on Hotels | hotels.com
Ad www.hotels.com/Amsterdam/Best-Hotels ▼
Book the Best Hotel in Amsterdam. Price Guarantee, No Booking Fee. Exclusive Deals. Last Minute Hotel Deals. Guest Reviews. Travel Guides. No Cancellation Fees. Photos & Reviews. Luxury Hotels. Budget Hotels. Earn Free Hotel Nights. Types: Hotel, Apartment, Hostel, B&Bs, Holiday Park.

Ads

Search Engine Advertising (SEA)



The 30 best hotels & places to stay in Amsterdam, Netherlands ...
<https://www.booking.com> › Netherlands › Noord-Holland › Visit Amsterdam ▼
Great savings on hotels in Amsterdam, Netherlands online. Good availability and great rates. Read hotel reviews and choose the best hotel deal for your stay.

30 Best Amsterdam Hotels, Netherlands (From \$16) - Booking.com
<https://www.booking.com> › Netherlands › Noord-Holland › Visit Amsterdam ▼
Great savings on hotels in Amsterdam, Netherlands online. Good availability and great rates. Read hotel reviews and choose the best hotel deal for your stay.

THE 10 BEST Hotels in Amsterdam for 2019 (from \$62) - TripAdvisor
<https://www.tripadvisor.com> › ... › North Holland Province › Amsterdam ▼
The #1 Best Value of 1098 places to stay in Amsterdam. Free Wifi. Restaurant. XO Hotels Couture. Show Prices. #2 Best Value of 1098 places to stay in ...

Amsterdam Hotels | Find & compare great deals on trivago
<https://www.trivago.com> › Netherlands › North Holland ▼
Compare the prices of 6293 hotels in Amsterdam, Netherlands. Find the ideal rate from millions of accommodation deals and save with trivago.com.

Organic Search results

Search Engine Optimization (SEO)

SEA

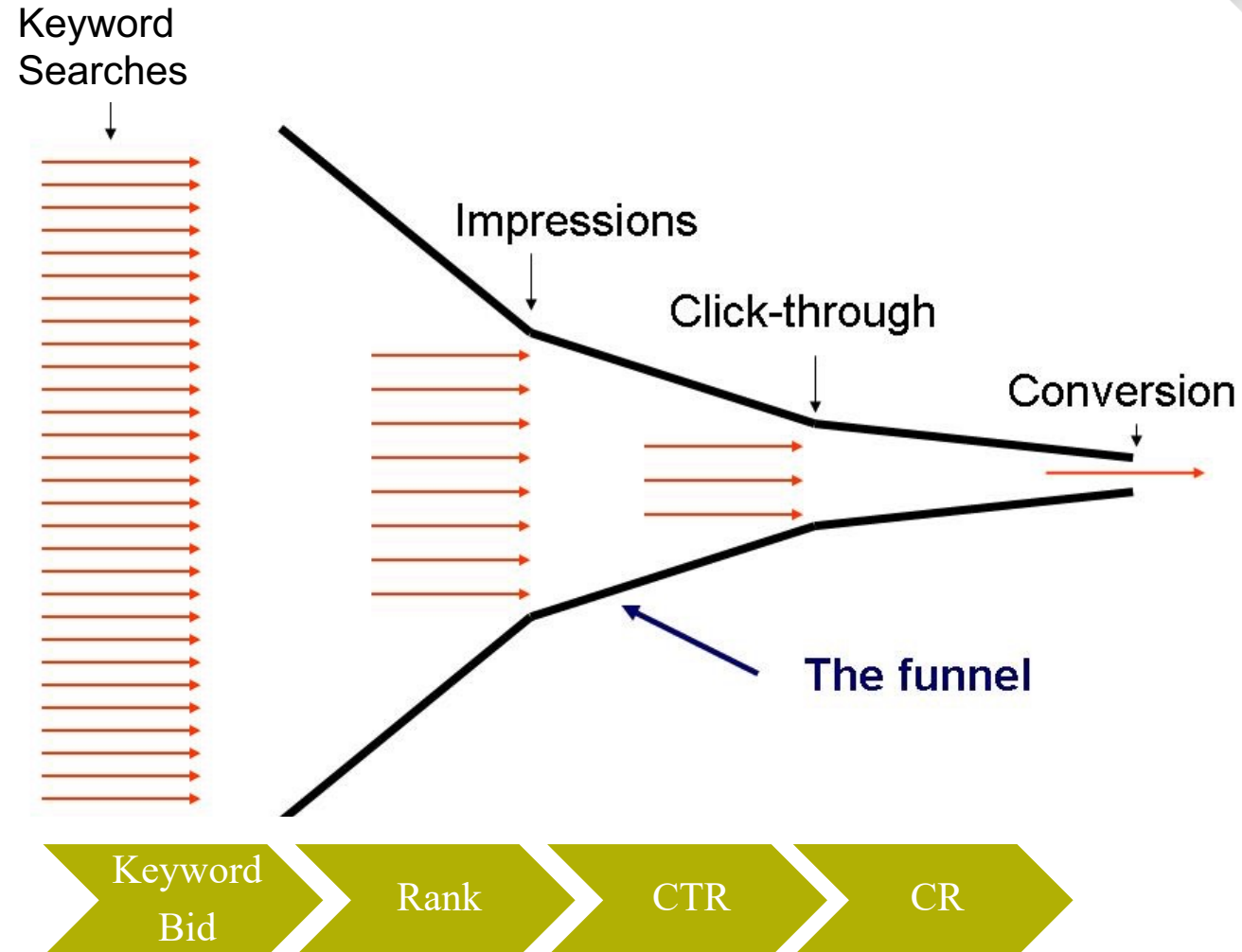


- Search engine **auctions** its advertisement space
- How much better is it to be #1 vs. #2?
- How much do I bid for someone searching “Hotel Amsterdam”?

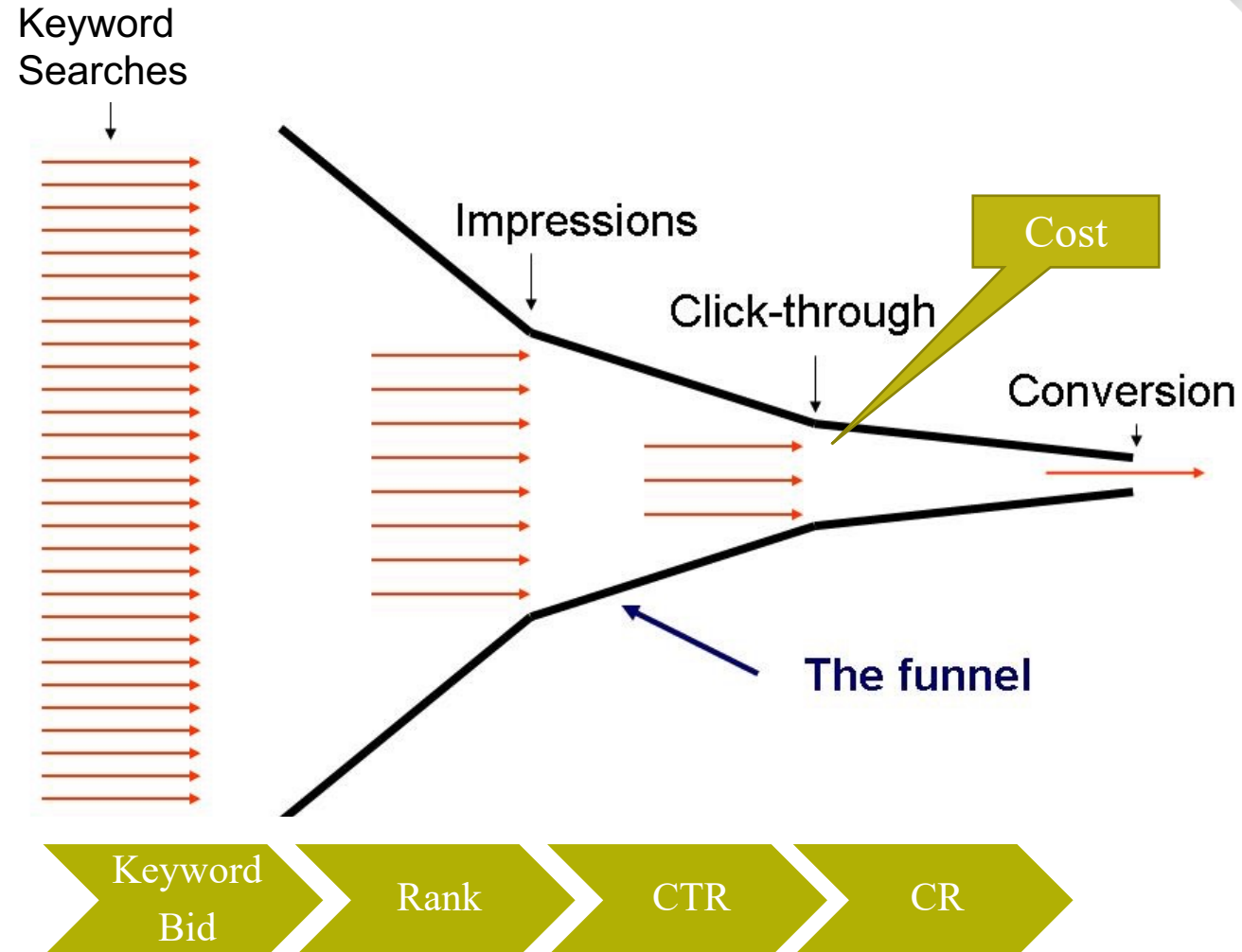
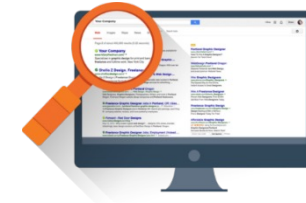
SEO

- Search engine displays results based on “relevance”
- How do I make my website relevant for the searched keyword?

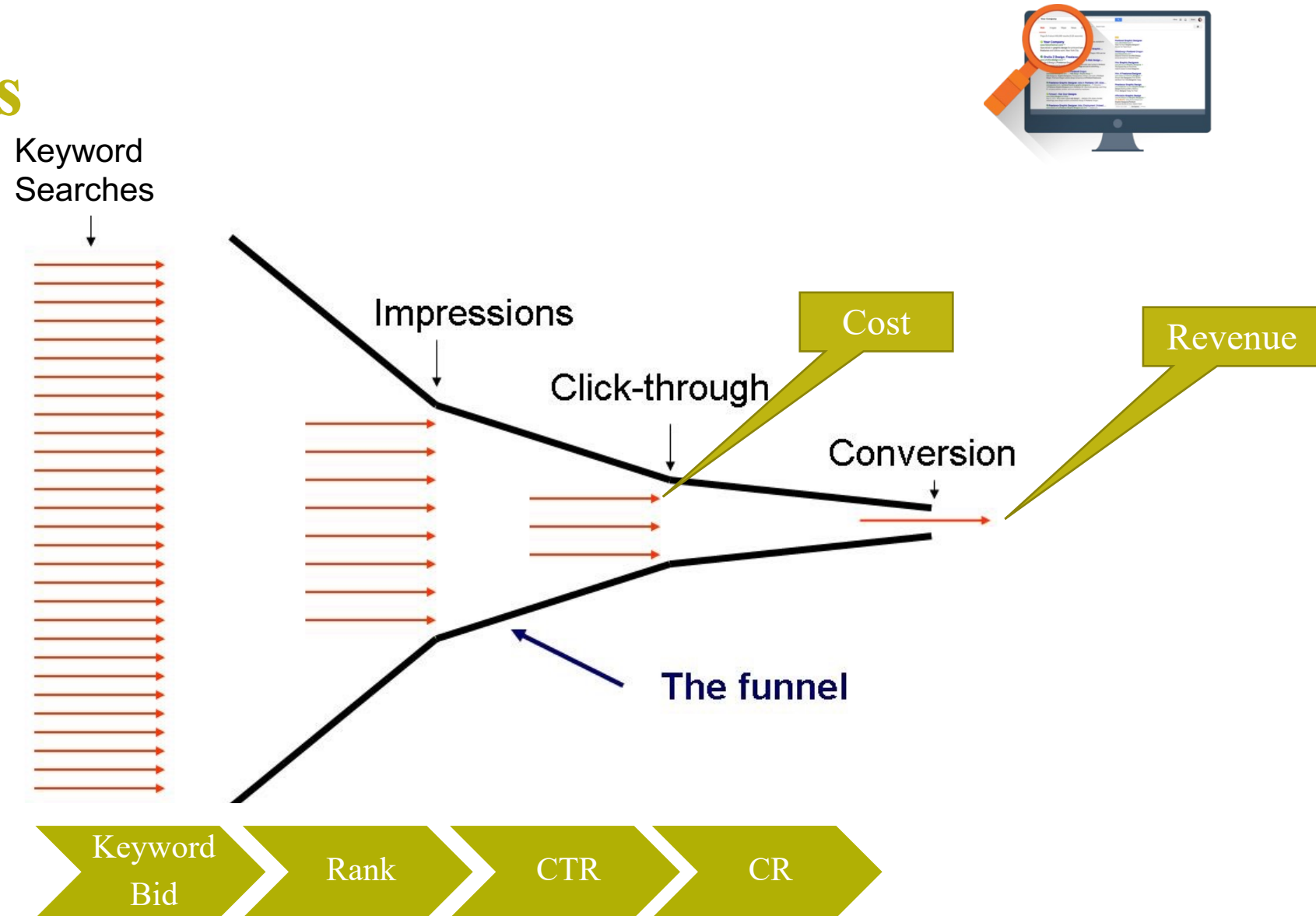
SEA process



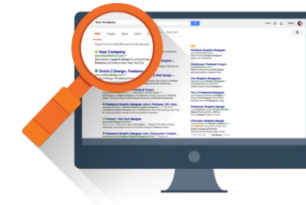
SEA process



SEA process



SEA process

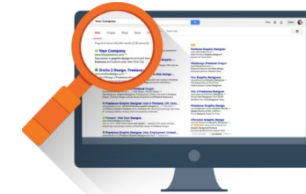
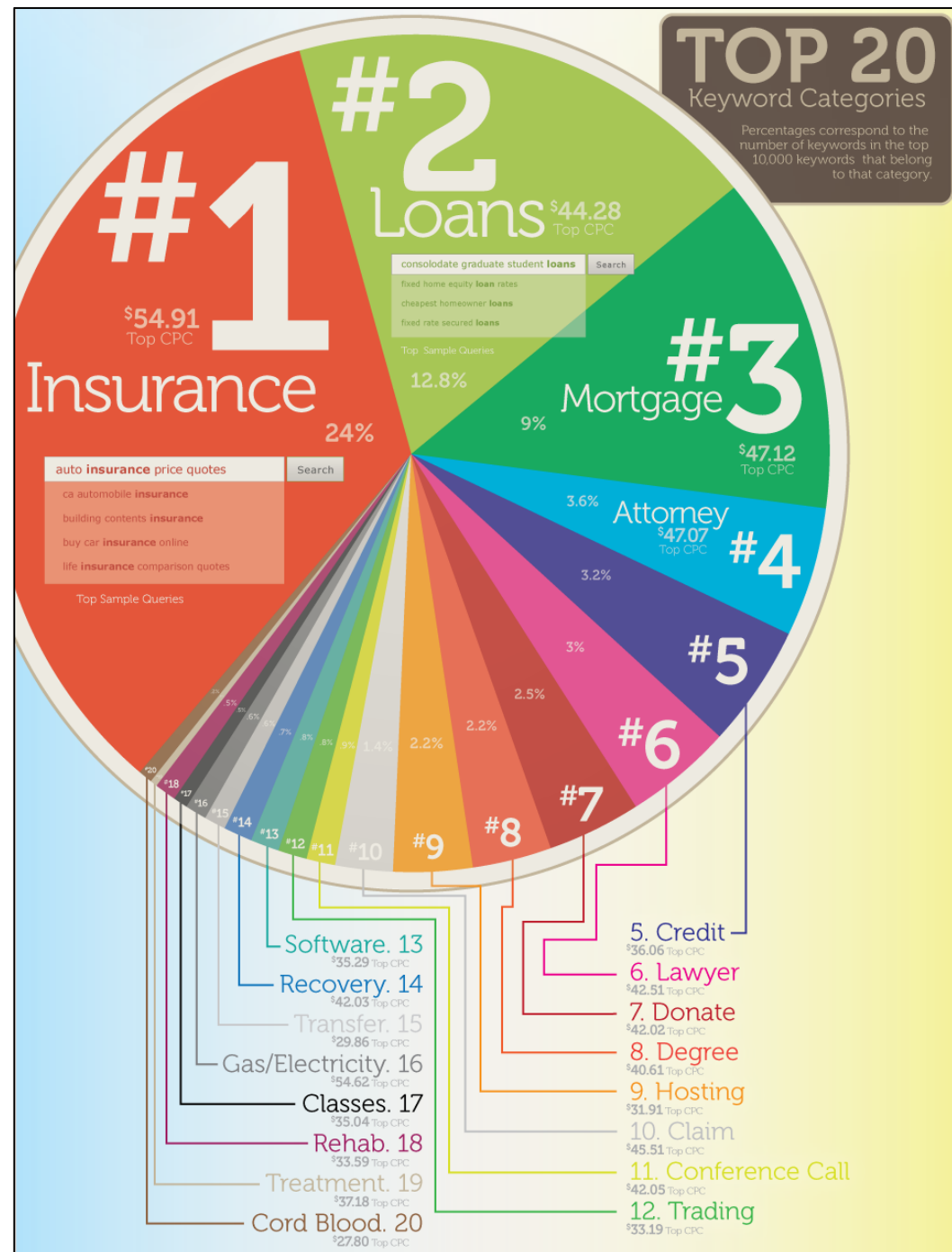


Advertisers

- create ads targeting keywords
- configure their bids (explicit or by setting goals and budgets)
 - A higher bid increases advertiser cost but also clicks and conversions.

Search Engine

- rank ads by bid and relevance (**click-through rate**)
 - Maximizes their revenue and relevance





Display





NU.nl navigation bar: Voorpagina, Economie, NUsport, Tech, Entertainment, Uit, Overig, Wonen, Auto, Shop, Mijn NU.nl

Your Ad Here

NU.nl logo | Vrijdag 25 januari 2019 | Het laatste nieuws het eerst op NU.nl

0 °C | 6 Files | Uit | TV gids | 0 Live

Voorpagina
Net binnen
Algemeen
Economie
Sport
Tech
Entertainment
Uit
Overig
Video's
Podcast
Regionaal
Adverteerders

Planbureau: Staat haalt klimaatdoelen niet, uitstoot zelfs toegenomen

Video: Italiaans dorp Gattolino overspoeld door 'gestoorde ratten'
Weer terugslag in reddingsoperatie Spaanse peuter, explosieven nodig
Vicepremier Italië wil ambassadeur Nederland op matje roepen
Graf ontdekkingsreiziger na tweehonderd jaar ontdekt in Londen
Hennenkwekerij ontdekt in geheime ruimte achter schilderij in huis Bergen

Net binnen

- 13:32 - Astrid de Jong genomineerd voor Nachtw...
- 13:20 - Adviseur Roger Stone van Amerikaanse p...
- 13:20 - Italiaans dorp Gattolino overspoeld door '...
- 13:19 - Uitjes in Nederland: Robotdino's en Zwols...
- 13:17 - Schorsing olympisch marathonkampioen...

Groene dromen?

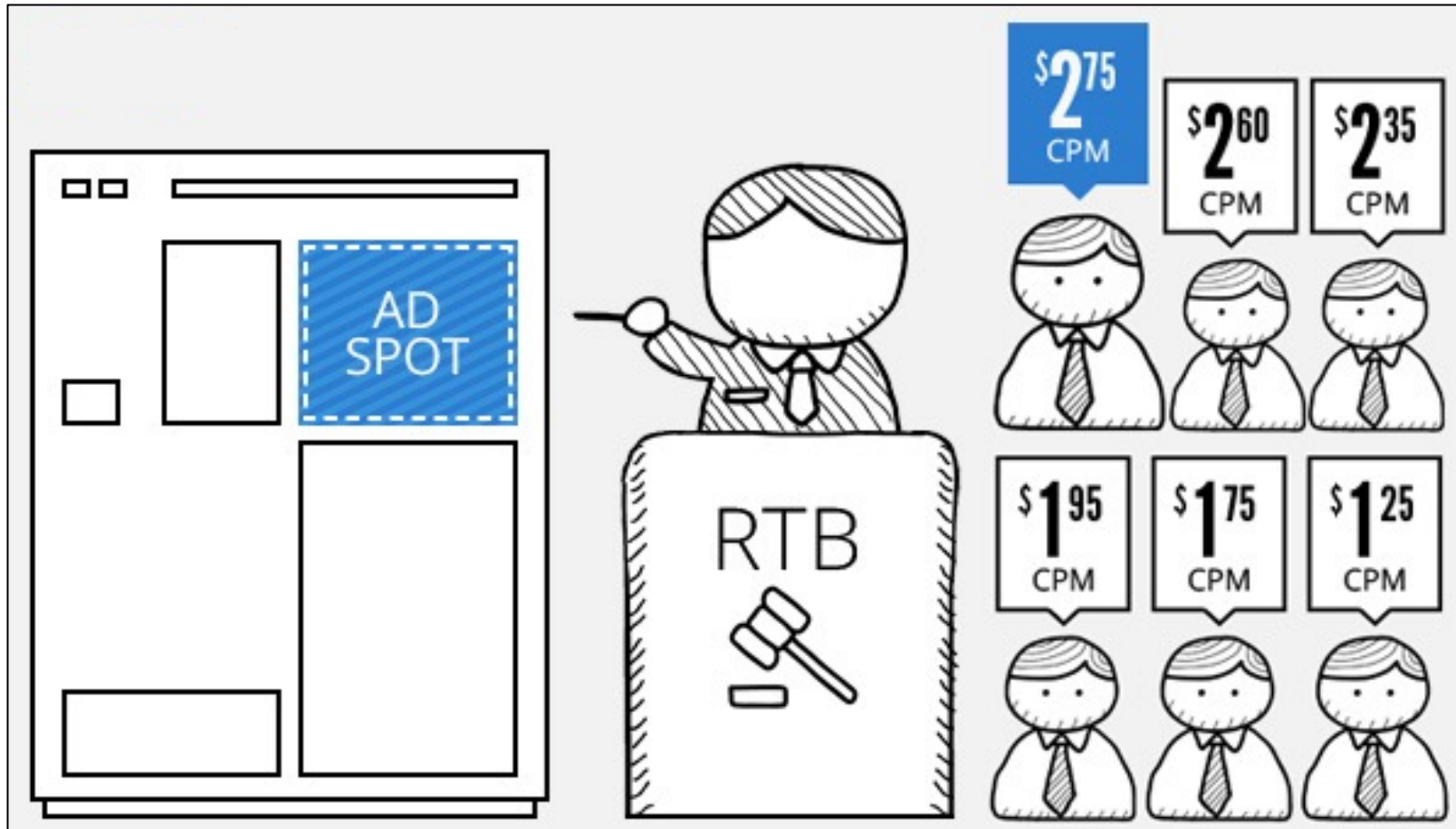
Kies voor lease, en start met groen doen.

Meer info

ABN-AMRO Lease



Display - Real-time Bidding (RTB)

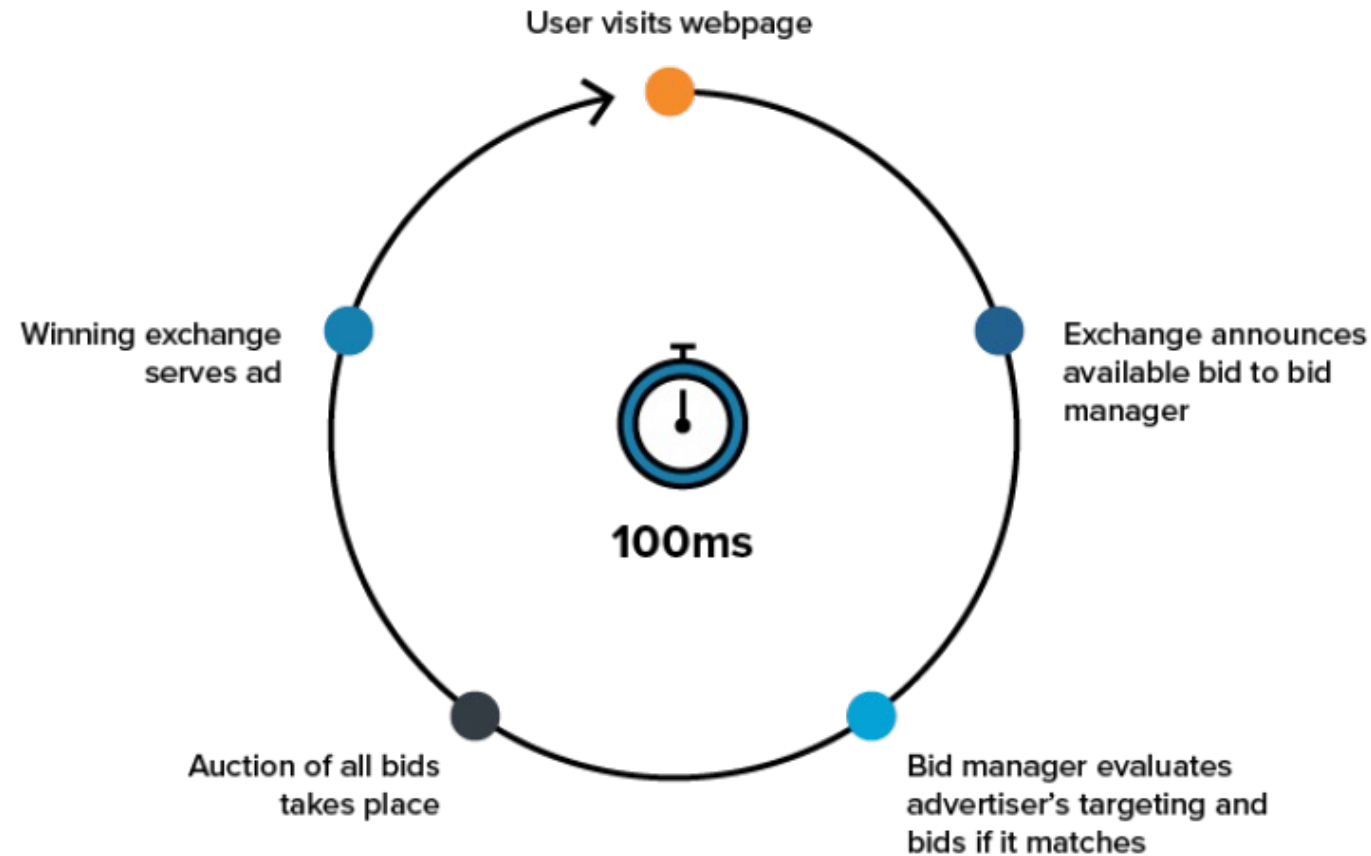


Real-Time Bidding (RTB)

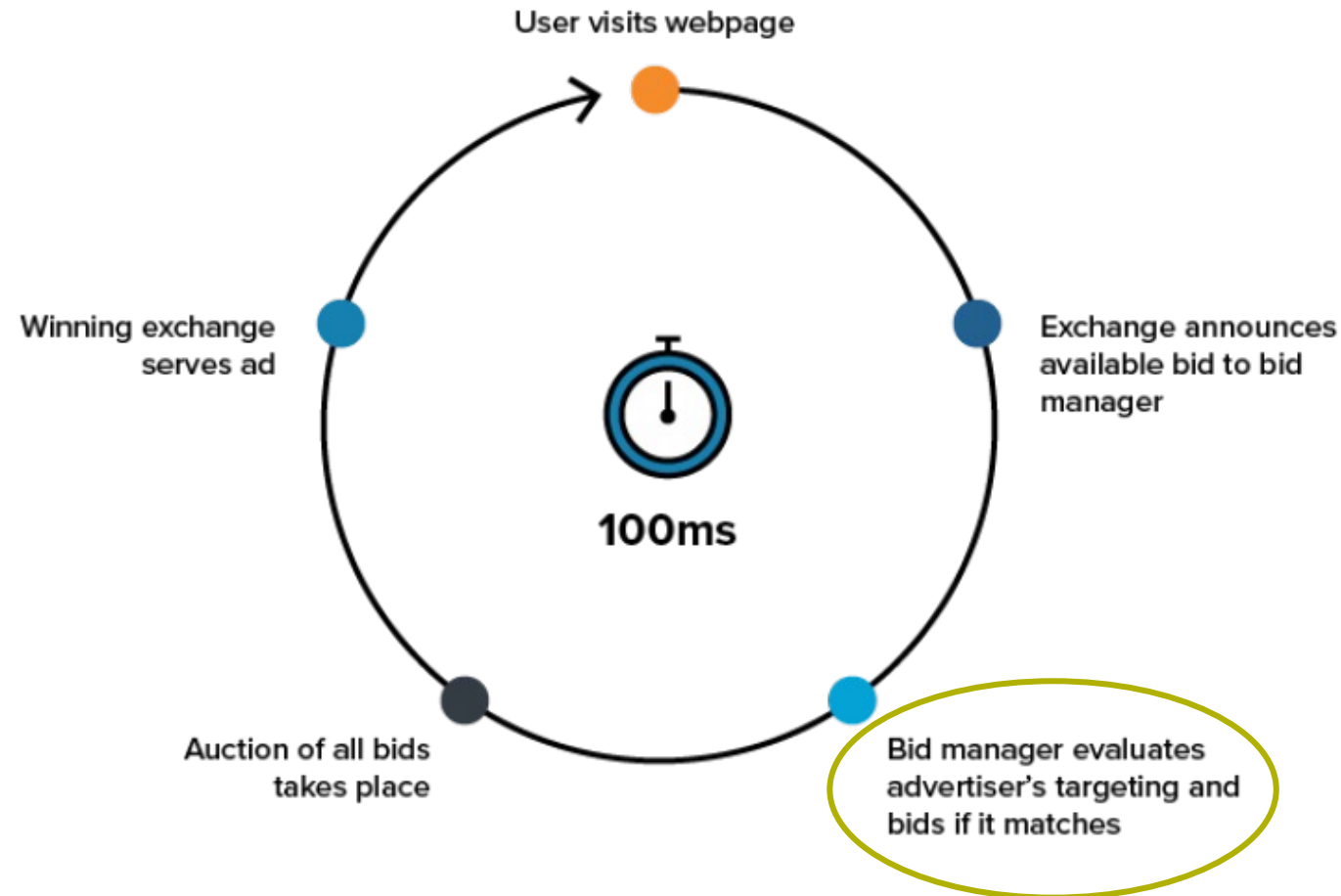


- RTB is more effective for both advertiser and publisher
 - More revenue for the publisher from the auction mechanism
 - Better audience targeting for the advertiser based on page, demographics, browsing history, location and time

Real-time Bidding (RTB) Process

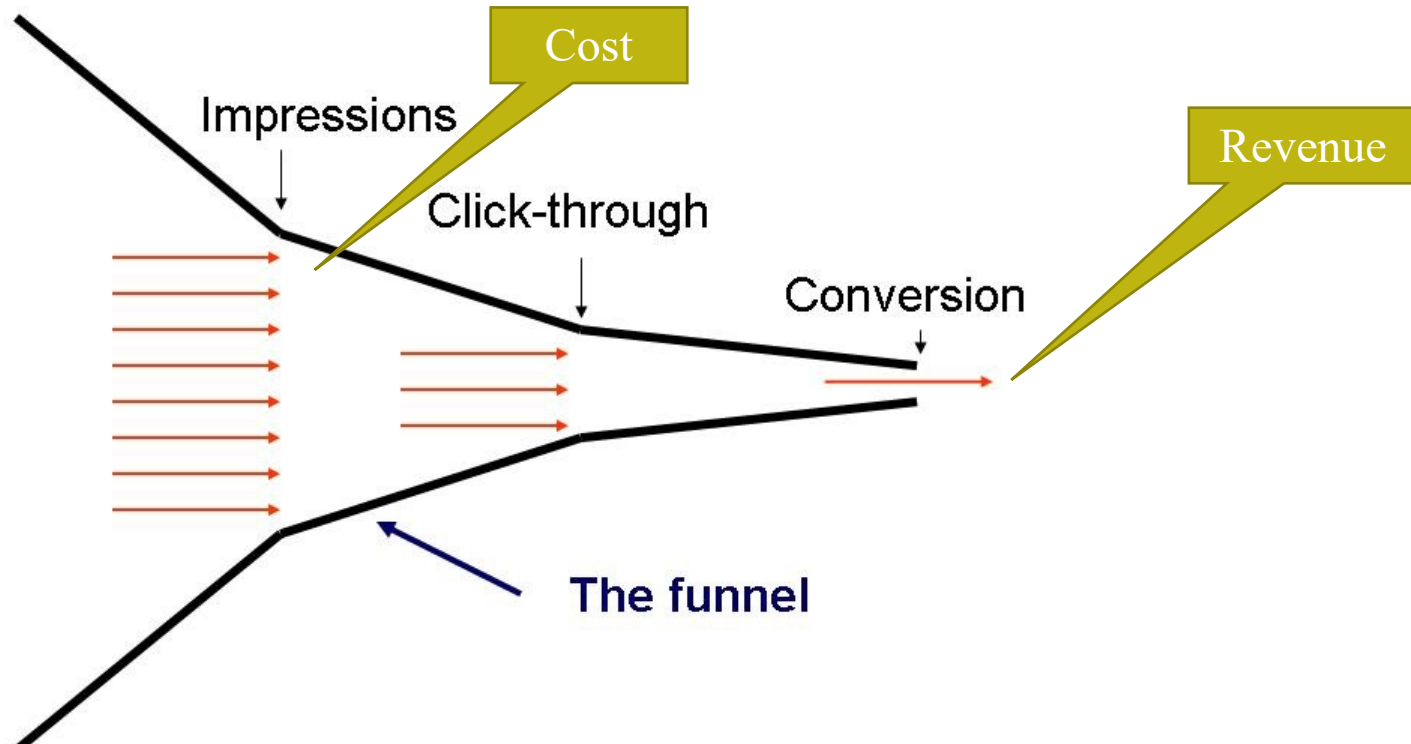
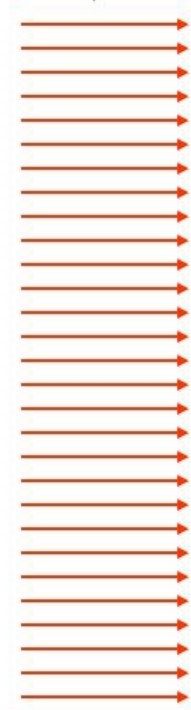


Real-time Bidding (RTB) Process



RTB process

Keyword
Searches



SEA vs. RTB



SEA

- Auction
- Multiple positions
- Targeting a keyword search
- Static bidding
- Cost per click
- “pull” marketing

RTB

- Auction
- Only 1 winning bid
- Targeting an individual page visit
- Real-time bidding
- Cost per view
- “push” marketing

Online Advertising - Outline

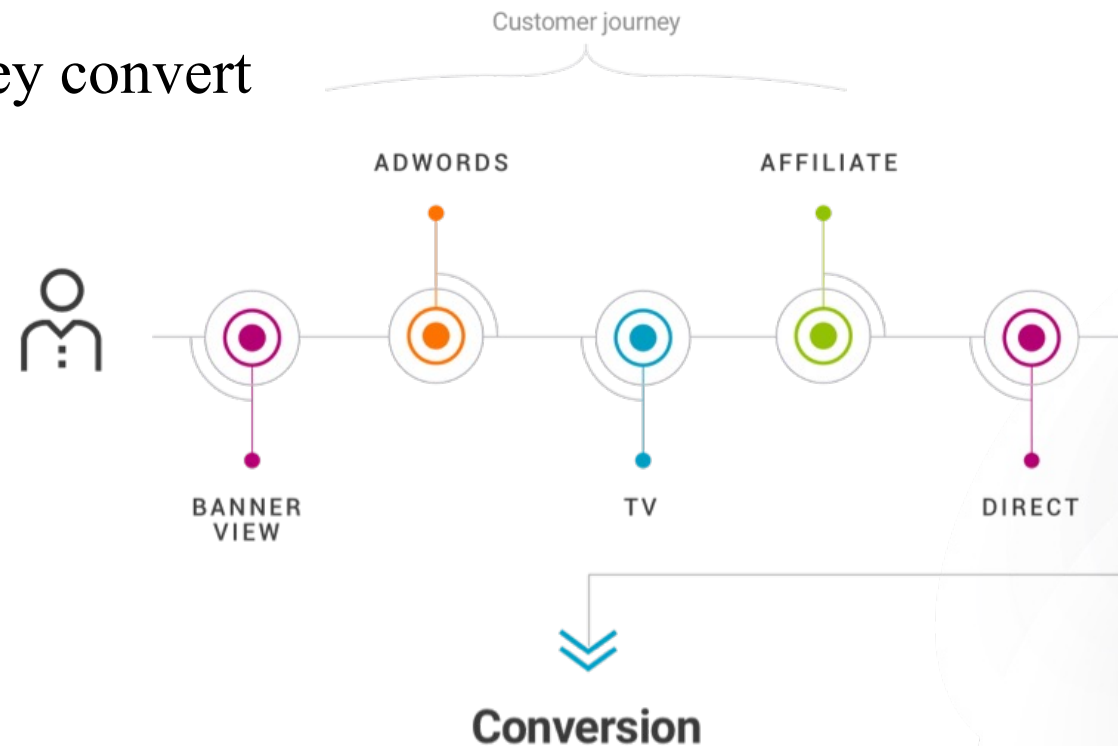
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 - ☐ Conversion attribution



Online Advertising – Multi Channel

Until now we've looked at optimizing individual channels, but ..

- advertisers use multiple channels at the same time
- people see multiple ads before they convert



Multichannel advertising modeling

Media Mix Modeling

Aggregate data on media spend and sales

“We advertised on TV last week and sales went up”

Conversion Attribution

Detailed data on customer touchpoints and conversions

“This person saw my advertisement on nu.nl yesterday and then entered my website. Today he came back and bought something”

Online, we have all the data, but...

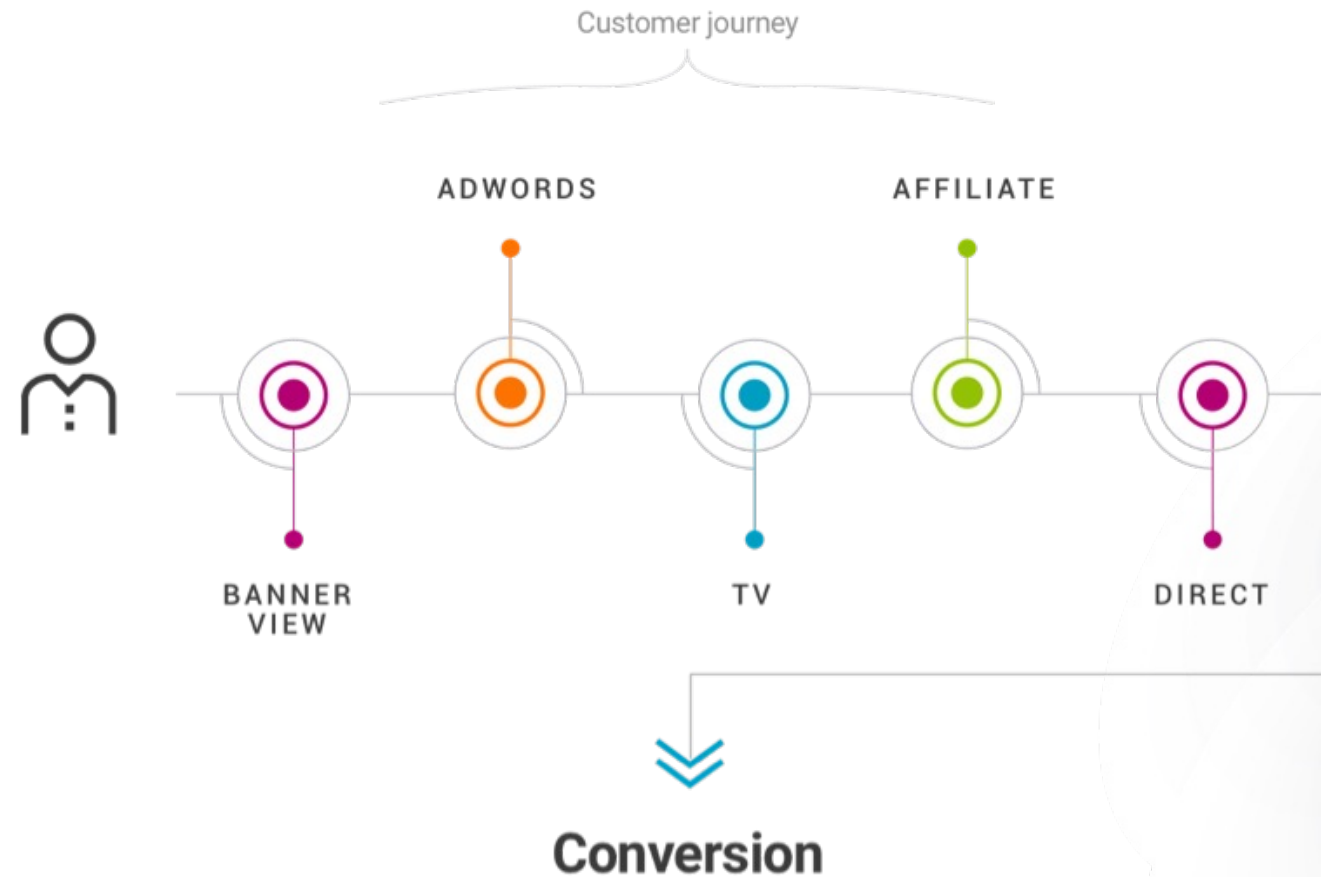
We have 3 ads:

- 2% of the people that click on our CNN ad convert
- 1.5% of the people that click on our Facebook ad convert
- 2.2% of the people that click on our Google ad convert

What about people that saw all 3?

- We're counting the same conversion multiple times
- How do we attribute the conversion over the 3 different channels?

Conversion Attribution



Industry Standard: Last Click Wins



far from logical

Next 2 weeks: CTR Modelling

- CTR modelling is at the heart of online advertising:
- Advertisers want to spend their money on good performing advertisements
- Publishers often get paid per click (PPC)

Reinforcement Learning	CTR Prediction
Learn on the fly which version of an ad has the best CTR. Balance exploration and exploitation in deciding which version to show next.	Predict CTR's where a large number of attributes are involved (about ad, impression, context, etc.).
Reinforcement learning models	Supervised Learning models that can handle a large number of attributes and can be updated on the fly.